

Teacher Guide — Electrostatics (Electric charge, Coulomb's law, electric field, Gauss's law)

Physics | Grade 11-12

Period 1

Entry Ticket

Look at the short scenario and write A or B and one short reason.

Scenario: A metal doorknob is touched by a small charged metal sphere. Where is the extra charge after contact?

A: The extra charge stays only where the sphere touched the knob.

B: The extra charge spreads over the knob (and other metal parts connected to it).

Choose A or B and write one short reason for your choice.

Expected response: Correct answer: B. Reason: In conductors (metal doorknob) charge moves freely and redistributes over the metal surface. A is a common misconception that comes from seeing charge stay where it was placed on insulators.

Teacher Script

0-5 min: Hook: quick demonstrations and words

Hold up a plastic comb and rub it on wool (or use pre-charged plastic strip). Say: "Watch this comb and the small paper bits." Drop a few small bits of paper and bring the comb near so they jump. Then snap a finger near a metal doorknob (or show a recorded short spark) and say: "Sometimes we see a little spark when we touch a handle." Ask: "What force did you see? Was something attracted or repelled? How strong did it feel—tiny, small, or big?" Give students 1 minute to sketch or write one line describing direction (toward/away) and strength (tiny/small/big). Collect two quick answers aloud.

Board: Write short labels as students answer: 'attracted / repelled' and 'tiny / small / big' in two columns.

- Anticipate: Why did the paper jump to the comb?
- Anticipate: Why do I sometimes get a spark and sometimes not?
- Anticipate: Was that the same thing when my hair stood up?

5-15 min: Introduce charge types and conductor vs insulator

Say: "We call the property that makes the paper jump and the spark 'electric charge.' There are two kinds of charge. Like charges push away; unlike charges pull together. I'll show you with this gold-leaf electroscope and a charged rod." Demonstrate: touch a charged plastic rod to the electroscope knob and show the leaves diverge. Say: "When the charged object touched the knob, charge moved to the leaves and they separated. That shows the metal lets charge move." Then hold a small plastic bead charged and place it on a plastic plate (insulator) and point: "On the plastic plate, the mark where I put the charge stays local—it does not spread." Read the exact definition on the board for insulator: "A substance that offers high resistance to the passage of electricity and retains charge where it is placed." Emphasize: conductors let electrons move and redistribute; insulators keep charge where put.

Board: Write heading 'Two kinds of charge' and below it write 'Like repel, unlike attract'. Under that write the definition line exactly as given for insulator. Draw two small sketches: (1) metal sphere + leaves diverging, arrow showing charge moving; (2) plastic plate with a dot showing localized charge.

- Anticipate: What do we mean by 'two kinds'—are they called positive and negative?
- Anticipate: Is metal the only conductor?

- Anticipate: If I put charge on plastic sometimes it spreads a little—why?

15-27 min: Derive Coulomb idea from experiment (no formula written yet)

Say: "We want a rule for the force between two small charged objects. First we idealize each object as a point charge if they are small compared to their separation. Watch this simple experiment: two small charged pith balls hung on strings; measure how force changes when we double the distance." Place two labeled pith balls hanging so they repel. Ask: "If I move them twice as far apart, how does the repulsion change?" Collect predictions. Then say: "Empirically, the force drops a lot when distance increases; experiments show it falls off with the square of the distance between the point charges ($1 \text{ over } r \text{ squared}$)." Point to the pith balls and say: "We introduce a constant k to match units so that the force has units of newtons (N). The magnitude of the force depends on the size of both charges, the separation r , and a proportionality constant k . Direction: if charges have the same sign the force is along the line joining them and points outward (repel); if opposite sign it points inward (attract)." Emphasize units: talk through an example verbally: "If q_1 is a few coulombs (C) and q_2 is a few C and r in metres (m), then the force we measure is in newtons (N). We will write and use the precise formula next period."

Board: Write headings: 'Point-charge idealisation', 'Empirical: force falls with distance squared ($1/r^2$)', 'Introduce constant k to give units of N'. Draw two point dots labeled q_1 and q_2 with a line and label r and arrows showing 'repel' or 'attract' depending on sign.

- Anticipate: Why can we treat objects as point charges?
- Anticipate: What does '1 over r squared' mean physically?
- Anticipate: What is k —can we compute it now?

27-35 min: Guided practice: calculate magnitude and direction (numbers, units)

Hand out or display the problem: 'Two point charges $q_1 = +2.0 \times 10^{-6} \text{ C}$ and $q_2 = -3.0 \times 10^{-6} \text{ C}$ are 0.50 m apart. Using the idea that force depends on q_1 , q_2 and $1/r^2$, determine the direction of the force on q_1 and state units you expect for force.' Give pairs 5 minutes to work. Circulate and listen for unit work: check that they keep coulombs for charge and metres for distance and that force is reported with units newtons (N). If a pair computes a magnitude now, encourage them to state it as 'some constant $\times (2.0 \times 10^{-6} \text{ C} \times 3.0 \times 10^{-6} \text{ C}) / (0.50 \text{ m})^2$ ' and then say 'result in newtons' — don't require the numeric value of k today. After 5 minutes, ask one pair to state the direction of the force on q_1 : attract (toward q_2) because signs are opposite.

Board: Write the problem statement exactly as given. Underline q_1 , q_2 and r . Write sample setup: 'magnitude $(|q_1||q_2|)/(r^2)$, units: $(\text{C} \times \text{C})/(\text{m}^2)$ N after constant k introduced'. Write answer for direction: ' q_1 feels force toward q_2 (attraction) because signs opposite.'

- Anticipate: Why did you use absolute values for magnitude?
- Anticipate: Do we multiply the charges or add them?
- Anticipate: When do we include the sign—does sign change magnitude?

35-40 min: Check understanding and link forward

Give students a short written diagnostic with two parts: (1) A positive charge placed on a metal sphere: does the charge stay where placed or spread? Write one sentence. (2) Units question: If q has units coulombs (C) and distance is metres (m), list the units that appear in the ratio $(q_1 \times q_2)/r^2$ before we insert any constant. Collect answers quickly by show of hands for (1) and scan a couple of students' written (2). Then say: "Next period we will write the exact formula with the constant and learn how to add forces from several charges (superposition) and why charge comes in discrete amounts (quantisation)."

Board: Write the two diagnostic questions on the board for students to copy. Below write: 'Next: exact formula,

superposition, quantisation'.

- Anticipate: Do we ever measure k in the lab?
- Anticipate: Will we calculate actual force numbers next time?
- Anticipate: What does quantisation mean for these charges?

Blackboard Notes

- Observed: attracted / repelled | tiny / small / big
- Two kinds of charge
- Insulator (exact definition)
- Point-charge idea and empirical law (no formula today)
- Practice problem
- Quick check / Next
- Two kinds of charge: like charges repel, unlike charges attract
- Insulator: A substance that offers high resistance to the passage of electricity and retains charge where it is placed.
- Conductors: charge (electrons) moves freely and redistributes over the surface
- Point-charge idealisation: treat small charged objects as points if size $\ll$ separation
- Empirical: force falls with distance squared ($1/r^2$) — magnitude depends on both charges and on $1/r^2$
- Introduce constant k to make force have units of newtons (N) — exact formula written next period
- Practice: $q_1 = +2.0 \times 10^{-6} \text{ C}$, $q_2 = -3.0 \times 10^{-6} \text{ C}$, $r = 0.50 \text{ m}$ direction on q_1 : toward q_2 (attraction)
- Units check: $(q_1 \times q_2)/r^2$ has units $(\text{C} \times \text{C})/(\text{m}^2)$ before including k
- Diagram: Comb picking up small paper bits; arrow showing attraction
- Diagram: Gold-leaf electroscope: metal knob, leaves diverging after contact; arrows showing charge moving to leaves
- Diagram: Plastic plate with a dot marking localized charge (insulator)
- Diagram: Two point charges q_1 and q_2 separated by line labeled r ; arrows along line showing 'repel' (same signs) and 'attract' (opposite signs)
- Diagram: Two pith balls hanging on strings showing repulsion (angles shown) to illustrate experiment

Activities

Compute the Coulomb Force Between Two Point Charges (12 min, experiment)

Materials: One printed slip per pair with values for q_1 , q_2 (in coulombs) and separation r (in metres), Scientific calculator per pair, Ruler (metric) or tape measure, Pencil and scrap paper, Whiteboard or chart paper and marker (teacher)

1. Arrange students in pairs. Hand each pair one printed slip containing two point charges q_1 and q_2 and their separation r . Tell students they have 12 minutes to derive Coulomb's law and calculate the force magnitude and direction for their numbers.
2. At the start say: 'We will derive Coulomb's law from the proportionality idea, then use it to compute a numerical force with units. Use $k = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$.' Write on the whiteboard the three-step derivation outline: (a) force magnitude is proportional to $|q_1 q_2|$, (b) it decreases with the square of the distance divide by r^2 , (c) combine with constant k to give $F = k |q_1 q_2| / r^2$ with units N. Underline units of each quantity as you say them: coulomb (C),

metre (m), newton (N).

- 3. While pairs work, circulate. For any pair that substitutes numbers incorrectly, point to the units and ask: 'What are the units of $q_1 \cdot q_2$ and of r^2 ? How do they produce newtons when multiplied by k ?' If a pair is stuck setting direction, prompt them to draw the two charges on paper and decide attraction versus repulsion depending on sign.
- 4. After 9 minutes, ask pairs to stop calculations and prepare a one-sentence result on their paper giving the force magnitude in newtons, the direction (toward or away), and a short reason based on charge signs. Collect three volunteers to read results in the final 2-3 minutes and correct any arithmetic or unit mistakes on the board.

Student instructions:

- 1. Read the slip your teacher gave your pair. Record q_1 (in C), q_2 (in C), and r (in m).
- 2. Derive the form of Coulomb's law step by step on your scrap paper: say $F \propto |q_1 q_2|$, then $F \propto 1/r^2$, combine to write $F = k \cdot |q_1 q_2|/r^2$. Label k with units: $k = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$. Show units: $\text{C} \cdot \text{C}$ gives C^2 ; r^2 gives m^2 ; k cancels m^2/C^2 to give N.
- 3. Substitute the numerical values from your slip into $F = k \cdot |q_1 q_2|/r^2$. Compute the magnitude using your calculator and write the answer with units N to two significant figures (or as appropriate).
- 4. Determine direction: draw two charges on your paper, write their signs, then state whether the force on q_1 is toward q_2 (attraction) or away from q_2 (repulsion). Write the force as a vector statement such as ' $F_{\text{on } q_1} = 3.4 \times 10^{-3} \text{ N}$ toward q_2 along the line joining them.'
- 5. If either q is zero or very small, note that the force will be zero or very small and record that with units. Prepare one sentence with your final magnitude, units, and direction to report.

Success looks like: Each pair writes the algebraic form $F = k \cdot |q_1 q_2|/r^2$ on their paper and labels k with its units $\text{N} \cdot \text{m}^2/\text{C}^2$; Each pair shows a numerical substitution with units carried through (C, m) and computes a magnitude in newtons on their calculator; Each pair has a small sketch of the two charges and can point to whether the force is toward or away from the other charge, with that direction written in words on their paper.

Support: If you are stuck, use this sentence frame and worked first item. Frame: ' F is proportional to $|q_1 q_2|$ and inversely proportional to r^2 , so $F = k \cdot |q_1 q_2|/r^2$. Substituting $k = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$, $q_1 = __ \text{ C}$, $q_2 = __ \text{ C}$, $r = __ \text{ m}$ gives $F = 8.99 \times 10^9 \times (|q_1 q_2|) / r^2 = __ \text{ N}$.' Worked example: If $q_1 = +2.0 \times 10^{-6} \text{ C}$, $q_2 = -3.0 \times 10^{-6} \text{ C}$, $r = 0.20 \text{ m}$ then $|q_1 q_2| = 6.0 \times 10^{-12} \text{ C}^2$, $r^2 = 0.040 \text{ m}^2$, so $F = 8.99 \times 10^9 \times 6.0 \times 10^{-12} / 0.040 = 1.35 \text{ N}$ toward q_2 (because charges are opposite). Pair lower-reading students with a stronger reader for calculation checking.

Extension: Give the pair a third charge q_3 at a different position on the same line. Ask them to compute the net electrostatic force on q_1 from q_2 and q_3 by calculating both forces with units, assigning directions (signs), and adding them vectorially to produce a single net force magnitude and direction in newtons. Explain how superposition leads to the net force.

Checkpoint Questions

- [apply] Two small point charges sit 0.50 m apart. $q_1 = +2.0 \times 10^{-6} \text{ C}$ and $q_2 = +3.0 \times 10^{-6} \text{ C}$. Without doing a long derivation, state whether the force on q_1 due to q_2 is toward q_2 or away from q_2 , and give the sign you would attach to that force component on q_1 (positive or negative along the line from q_1 to q_2).

Expected: Both charges are positive so they repel. The force on q_1 is away from q_2 (i.e., along the line from q_1 toward increasing separation). If we take the direction from q_1 to q_2 as positive, the force on q_1 is negative (points opposite that chosen positive direction).

- [understand] A charged metal sphere is touched to a neutral conducting pith-ball; afterwards the pith-ball moves away from the sphere. What does this tell you about how charge behaved

on the conductor and whether the pith-ball and sphere have like or unlike net charges? Keep your answer short and specific.

Expected: Touching a conductor lets charge move onto the pith-ball, so charge redistributed and some transferred to the pith-ball. The pith-ball moves away, so the net charges on the sphere and pith-ball are like (same sign), causing repulsion.

- [apply] You are given $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$ for calculations. Compute the magnitude of the force between $q_1 = 1.0 \times 10^{-6} \text{ C}$ and $q_2 = -2.0 \times 10^{-6} \text{ C}$ separated by $r = 0.20 \text{ m}$. Show units in the final numerical answer.

Expected: Calculate magnitude using the Coulomb expression (as derived in class): $F = k |q_1 q_2| / r^2$. Plugging values gives $F = (8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2) \times |(1.0 \times 10^{-6} \text{ C})(-2.0 \times 10^{-6} \text{ C})| / (0.20 \text{ m})^2 = (8.99 \times 10^9) \times (2.0 \times 10^{-12} \text{ C}^2) / 0.04 \text{ m}^2 = (17.98 \times 10^{-3} \text{ N}\cdot\text{m}^2 / \text{m}^2) = 0.01798 \text{ N} \approx 1.8 \times 10^{-2} \text{ N}$. Units: newtons (N).

Exit Ticket

Two point charges are fixed on a horizontal line: $q_A = -4.0 \times 10^{-6} \text{ C}$ at $x = 0.00 \text{ m}$ and $q_B = +1.0 \times 10^{-6} \text{ C}$ at $x = 0.30 \text{ m}$. A third small test charge $q_T = +2.0 \times 10^{-7} \text{ C}$ is placed at $x = 0.10 \text{ m}$. Using $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$, calculate (a) the magnitude and direction of the net electrostatic force on q_T (give direction along the line: left or right) and (b) show the unit for your final force.

Success indicator: Correct net force value (within 5%) with units N, and correct direction (left or right). Work shows separate forces from q_A and q_B on q_T , their vector (left/right) signs, numeric addition, and the final unit 'N'.

Homework

- Calculate the magnitude and direction of the force between each pair for these three cases. Use $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$ and show units in every step: (a) $q_1 = +5.0 \times 10^{-7} \text{ C}$ and $q_2 = +5.0 \times 10^{-7} \text{ C}$ separated by 0.10 m , (b) $q_1 = -3.0 \times 10^{-6} \text{ C}$ and $q_2 = +1.0 \times 10^{-6} \text{ C}$ separated by 0.25 m , (c) $q_1 = +2.0 \times 10^{-8} \text{ C}$ and $q_2 = -6.0 \times 10^{-8} \text{ C}$ separated by 0.05 m .
- Explain in one short paragraph (3–4 sentences) why a conductor and an insulator behave differently when you place charge on them; include the definition of 'insulator' from class and mention electrons.
- Sketch a labelled diagram for the class electroscope demonstration that shows charge transfer when a charged object touches the knob and explain in one sentence how the leaf divergence indicates the presence of net charge.

Estimated time: 30 min

Mentor Moment — The Balloon That Stuck to the Wall

I once rubbed a balloon on my sweater and it stuck to a classroom wall for several minutes. The balloon had gained charge while the wall stayed (mostly) neutral; the attraction came from the wall's charges slightly shifting toward the balloon.

Takeaway: Rubbing transfers charge; conductors let charge move, but even neutral insulators can be attracted by a charged object because charges within them shift slightly (induced alignment).

Period 2

Entry Ticket

Quick: I have two small plastic spheres. I rub one and it becomes negatively charged, and the other becomes positively charged. Do you think I created new charge, or moved existing charge between the spheres? Give one sentence and one short reason.

Expected response: I moved existing charge between the spheres; rubbing transfers electrons from one object to the other, so net charge of the pair is unchanged.

Teacher Script

0-5 min: Warm-up and recall

Read the entry-ticket answers aloud: "Did rubbing create charge, or move it?" Ask three students to state their one-sentence answer. If a student says 'created', ask: "What would conservation mean for the two-sphere system after rubbing? Would net charge be zero, positive or negative?" Show the previous-period schematic of two point charges (one +, one -) placed 0.10 m apart. Ask the class: "Which way will the force on the positive charge point?" Call on a student to point to the direction. Repeat quickly the terms conductor/insulator and remind where charges live in atoms (electrons, protons).

Board: Write two quick bullets: 1) Rubbing transfers electrons; 2) Net charge conserved. Draw the two-point-charge schematic (label + and -, 0.10 m).

- Anticipate: If rubbing moves electrons, where do the electrons come from?
- Anticipate: Can rubbing ever create a tiny new electron?
- Anticipate: How do we know the force direction without calculating?

5-12 min: Quantisation and conservation of charge

Say: "Charge comes in discrete units. One electron has charge $-e$, one proton has $+e$, where $e = 1.60 \times 10^{-19}$ C." Write $q = n e$. Then show a concrete example: a neutral plastic rod picks up 5×10^{12} extra electrons when rubbed. Ask: "What is the rod's charge in coulombs?" Walk the class through multiplication: $n = 5 \times 10^{12}$, $e = 1.60 \times 10^{-19}$ C, so $q = (5 \times 10^{12})(-1.60 \times 10^{-19} \text{ C}) = -8.0 \times 10^{-7}$ C. Emphasize carrying units: write each step with C. Next show a two-object transfer example: if object A loses 2.0×10^6 electrons and object B gains them, what are final charges? Do the arithmetic together and point out net charge change is zero. Explicitly correct the rubbing-makes-charge idea: say "We observe charge appearing on one object and disappearing on another — transfer, not creation."

Board: Write formula $q = n e$. Under it, calculation lines: $n = 5 \times 10^{12}$; $e = 1.60 \times 10^{-19}$ C; $q = n e = (5 \times 10^{12})(-1.60 \times 10^{-19} \text{ C}) = -8.0 \times 10^{-7}$ C. Then show short transfer example: A loses $2.0 \times 10^6 e$ $q_A = -(2.0 \times 10^6)(1.60 \times 10^{-19} \text{ C})$ etc., and total remains zero.

- Anticipate: Why is e the same for all electrons?
- Anticipate: Are there charges smaller than e ?
- Anticipate: How do we count such huge numbers of electrons in practice?

12-22 min: Derive Coulomb's law (statement and vector form)

Say: "Experiments show force between point charges falls like $1/r^2$ and is proportional to $q_1 q_2$." Draw two point charges q_1 and q_2 separated by r . Ask: "If I double r , what happens to the force?" (Students should say it becomes $1/4$.) Then write the scalar form: $F = (1/4\pi\epsilon_0) (q_1 q_2 / r^2)$. Define $1/4\pi\epsilon_0$ as k and state units: q in coulombs (C), r in metres (m), so F comes out in newtons (N). Now derive vector form step-by-step: 1) Define $\mathbf{r}_{21}$ = vector from charge 2 to charge 1 and $r = |\mathbf{r}_{21}|$. 2)

Define unit vector $\hat{r}_{21} = \mathbf{r}_{21}/r$. 3) State pairwise force on 1 from 2: $F_{12} = (1/4\pi\epsilon_0) (q_1 q_2 / r_{21}^2) \hat{r}_{21}$. Carry units through using example numbers: $q_1 = q_2 = 1.0 \mu\text{C} = 1.0 \times 10^{-6} \text{ C}$, $r = 0.10 \text{ m}$. Compute magnitude: $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$. Show multiplication: $F = (8.99 \times 10^9)(1.0 \times 10^{-6} \text{ C})(1.0 \times 10^{-6} \text{ C})/(0.10 \text{ m})^2 = (8.99 \times 10^9)(1.0 \times 10^{-12})/(0.01) = (8.99 \times 10^9)(1.0 \times 10^{-10}) = 8.99 \times 10^{-1} \text{ N} = 0.899 \text{ N}$. Say the direction: if both positive they repel along $\hat{r}$ from each other. Emphasize each step carries units and produces N.

Board: Draw two charges, label q_1 , q_2 and r . Write scalar Coulomb law: $F = (1/4\pi\epsilon_0) (q_1 q_2 / r^2)$. Below, write vector form: $F_{12} = (1/4\pi\epsilon_0) (q_1 q_2 / r_{21}^2) \hat{r}_{21}$. Then numeric example steps converting μC to C, write $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$, and the arithmetic to get $F = 0.899 \text{ N}$.

- Anticipate: Where does the $1/r^2$ come from physically?
- Anticipate: Why use a unit vector instead of writing the sign?
- Anticipate: What if one charge is negative — how do I get direction?

22-28 min: Demonstration / data from torsion-balance style setup

Say: "I'll show a short demo (or simulation clip) of a torsion-balance measurement of force vs distance." Run the demo or start the video. Ask students to watch for how force changes when distance doubles. After the clip, ask: "What pattern do you see for force vs r ?" Guide them to say it falls by about $4\times$ when r doubles. Show on board one pair of measured values from the demo: $r_1 = 0.05 \text{ m}$, $F_1 = 3.6 \text{ N}$; $r_2 = 0.10 \text{ m}$, $F_2 = 0.90 \text{ N}$. Ask them to check proportionality to $q_1 q_2 / r^2$ by dividing F by $(q_1 q_2 / r^2)$ using the demo's known charges and confirm near-constant k . Emphasize units (N for force).

Board: Write demo data: $r_1 = 0.05 \text{ m}$, $F_1 = 3.6 \text{ N}$; $r_2 = 0.10 \text{ m}$, $F_2 = 0.90 \text{ N}$. Compute ratio $F_1/F_2 = 4$ and show check $F \propto 1/r^2$. Note charges used in demo (copy values) and show one calculation dividing F by $(q_1 q_2 / r^2)$ to get $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$.

- Anticipate: How accurate are these demo numbers?
- Anticipate: Does the result change if the charges are not point-like?
- Anticipate: Why does the torsion balance measure force directly?

28-36 min: Guided practice — two problems in pairs

Give problem (a) aloud: "Compute magnitude and direction of force on $q_1 = +2.0 \times 10^{-6} \text{ C}$ from $q_2 = -3.0 \times 10^{-6} \text{ C}$ separated by $r = 0.20 \text{ m}$. Show units and sign/direction with unit vector." Pause and have pairs work for 3 minutes. Circulate and prompt students to convert numbers and use k . Then give problem (b): "Three charges on a line: $q_1 = +2.0 \times 10^{-6} \text{ C}$ at $x=0$, $q_2 = -1.0 \times 10^{-6} \text{ C}$ at $x=0.10 \text{ m}$, $q_3 = +1.0 \times 10^{-6} \text{ C}$ at $x=0.20 \text{ m}$. Find net force on q_2 . Use superposition and show vector sum with units." Let pairs continue for 5 minutes. Provide scaffolding: remind them to compute each pairwise force magnitude, assign directions with $+/-$ or unit vectors, then add.

Board: Write problem (a) and (b) exactly as stated. Under each, show the first step for (a): convert μC to C, write $F = k q_1 q_2 / r^2$, compute magnitude, then show direction: attraction toward q_2 . For (b) write pairwise forces F_{21} and F_{23} with signs and then $F_{\text{net}} = F_{21} + F_{23}$.

- Anticipate: Do I include a negative sign for attraction or use unit vectors?
- Anticipate: When adding forces on a line, do I treat right as positive?
- Anticipate: If two forces cancel partially, how do I show units?

36-40 min: Quick quiz and consolidation

Hand out a 2-question quiz (or show it on the board): Q1 (calculation): "Two point charges $+1.0 \times 10^{-6} \text{ C}$ and $+2.0 \times 10^{-6} \text{ C}$ separated by 0.15 m . Compute magnitude of force and state units." Q2 (concept): "If an isolated neutral object is rubbed and becomes $+3.2 \times 10^{-7} \text{ C}$, did rubbing create charge? Explain in one sentence mentioning electrons or protons." Give one minute to answer Q1

and one minute to answer Q2. Collect answers quickly (or have volunteers read). Give the correct brief answers: show numeric answer for Q1 and the conservation answer for Q2. End by stating: "Next time we use these point-source forces to define the Electric field and the idea of a test charge and source charge."

Board: Write the two quiz questions. After collection, write brief answers: numeric F for Q1 with units N, and the sentence 'charge transferred, not created; electrons moved' for Q2.

- Anticipate: Can we use Coulomb's law for charged spheres or only point charges?
- Anticipate: How small must a charge be to be a test charge?
- Anticipate: What if I get a different numeric answer using a calculator?

Blackboard Notes

- Quantisation of Charge
- Coulomb's Law
- Vector form and example
- Demo data (force vs distance)
- Practice problems
- Quiz
- Charge q is integer multiple of e: $q = n e$, $e = 1.60 \times 10^{-19} \text{ C}$
- Rubbing transfers electrons; net charge conserved in isolated system
- Scalar Coulomb law: $F = (1/4\pi\epsilon_0) (q_1 q_2 / r^2)$ (units: N when q in C, r in m)
- Vector form: $F_{12} = (1/4\pi\epsilon_0) (q_1 q_2 / r_{21}^2) \hat{r}_{21}$
- $k = 1/(4\pi\epsilon_0) = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$ (used in numeric examples)
- Example: $q_1 = q_2 = 1.0 \mu\text{C} = 1.0 \times 10^{-6} \text{ C}$, $r = 0.10 \text{ m}$ $F = 0.899 \text{ N}$ (repel if both +)
- Superposition: net force = vector sum of individual pairwise forces
- Diagram: Two point charges labeled q_1 and q_2 separated by r ; draw $\hat{r}$ from q_2 to q_1 and label r_{21}
- Diagram: Schematic of rubbing two spheres showing electrons moving from one to the other (arrow with e^-)
- Diagram: Torsion-balance style sketch with two distances r_1 and r_2 and corresponding forces F_1 and F_2
- Formula: $q = n e$
- Formula: $F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$
- Formula: $\mathbf{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{21}^2} \hat{r}_{21}$

Activities

Pairwise Coulomb Forces and Charge Quantisation Problems (10 min, problem_set)

Materials: printed worksheet (one per student) or whiteboard, calculator, pencil, ruler

1. Arrange students in pairs and hand out the worksheet. Say: "You have 10 minutes to complete 4 short problems about charge quantisation, Coulomb's law, and superposition. Show all units and steps." Start a visible 10-minute timer.
2. Read aloud the first problem so every pair hears one worked example: explain that $e = 1.60 \times 10^{-19} \text{ C}$ and that charges must be integer multiples of e . While pairs work, circulate and watch for the two misconceptions listed below. If a pair seems stuck after 3 minutes, give the scaffold in the support box verbatim.
3. At 6 minutes remaining, announce time checks and prompt pairs who finished to attempt the extension. Collect worksheets at the end of 10 minutes.

- 4. After collection, note any recurring errors (units, vector sign/direction, non-integer charge answers, rubbing-creates-charge statements) to address in the next whole-class debrief.

Student instructions:

- Work with your partner. Show all steps and include units in every calculation.
- Problem 1 (Quantisation): A small isolated object has net charge $+4.80 \times 10^{-19}$ C. Assume only electrons and protons carry charge. Determine (a) how many excess protons or missing electrons correspond to this net charge? State the integer number and justify why it must be an integer. (Use $e = 1.60 \times 10^{-19}$ C.)
- Problem 2 (Coulomb magnitude & direction): Two point charges lie on the x-axis. $q_1 = +3.2 \times 10^{-6}$ C at $x = 0.00$ m and $q_2 = -1.6 \times 10^{-6}$ C at $x = 0.10$ m. (a) Calculate the magnitude of the electrostatic force on q_2 due to q_1 . (b) Give the direction of that force as a unit-vector or as 'left/right' along the x-axis. Use $k = 8.99 \times 10^9$ N·m²/C².
- Problem 3 (Superposition): Add a third charge $q_3 = +3.2 \times 10^{-6}$ C at $x = 0.20$ m. Calculate the net electrostatic force (magnitude and direction) on q_2 from q_1 and q_3 . Show the vector addition step explicitly and include units (N).
- Problem 4 (Conservation and microscopic origin): A neutral metal sphere is struck by a charged rod and afterward has net charge -6.4×10^{-19} C. Explain in two sentences where that extra negative charge came from (name the particle) and why rubbing does not create charge but redistributes it. Use the idea of conservation of charge and quantisation in your answer.

Success looks like: Each pair writes an integer value for number of electrons/protons in Problem 1 and shows the division: net charge (C) $\div$ e (C) = integer.; For Problems 2 and 3 each pair shows numeric substitution into Coulomb's law with units (N·m²/C², C, m), computes magnitudes, and indicates direction on the x-axis (left or right or using $\pm x$ unit vector).; In Problem 3 each pair writes the two pairwise forces on q_2 separately, shows their vector sum (with signs), and gives the final magnitude and direction.

Support: If you are stuck start here: worked first item — Problem 1: net charge = $+4.80 \times 10^{-19}$ C. Divide by $e = 1.60 \times 10^{-19}$ C: 4.80×10^{-19} C $\div$ 1.60×10^{-19} C = 3. So the object has +3 elementary charges (three excess protons or three missing electrons). Sentence frame for Problem 4: "The extra negative charge is due to extra electrons (particle = electron). Rubbing does not create charge because charge is conserved; the electrons moved from ___ to ___, changing distribution but not the total number of electrons in the isolated system." If needed, pair a student who is unsure with the stronger reader in the pair.

Extension: A continuous line of three charges along the x-axis: $q_1 = +2.0 \times 10^{-6}$ C at $x = 0.00$ m, $q_2 = -1.0 \times 10^{-6}$ C at $x = 0.05$ m, $q_3 = -1.0 \times 10^{-6}$ C at $x = 0.15$ m. Calculate the net force on q_1 (magnitude and direction). Then explain whether the result would change if the signs of q_2 and q_3 were both flipped to positive, and why, using superposition and algebraic signs.

Checkpoint Questions

- [understand] Two small insulating spheres are rubbed so one ends up with $+3.2 \times 10^{-19}$ C and the other with -1.6×10^{-19} C. Explain in one sentence whether charge was created and give n for each sphere in $q = n e$ (use $e = 1.60 \times 10^{-19}$ C).

Expected: No charge was created; charge was transferred. For $+3.2 \times 10^{-19}$ C, $n = +2$ (because $3.2 \times 10^{-19} / 1.60 \times 10^{-19} = 2$). For -1.6×10^{-19} C, $n = -1$.

- [apply] Calculate the magnitude of the electrostatic force between $q_1 = +1.0$ μ C and $q_2 = +1.0$ μ C separated by $r = 0.10$ m. Show units and give the direction of the force on q_1 .

Expected: Convert charges: 1.0 μ C = 1.0×10^{-6} C. Use $F = (1/4\pi\epsilon_0) q_1 q_2 / r^2$ with $1/4\pi\epsilon_0 = 8.99 \times 10^9$ N·m²/C². $F = 8.99 \times 10^9 \times (1.0 \times 10^{-6})(1.0 \times 10^{-6}) / (0.10)^2 = 8.99 \times 10^9 \times 1.0 \times 10^{-12} / 0.01 = 8.99 \times 10^{-2}$ N = 0.0899 N. Direction: repulsive, so on q_1 the force is directly away from q_2 along the line joining them.

- [apply] Three charges lie on the x-axis: $+2.0\ \mu\text{C}$ at $x=0$, $-1.0\ \mu\text{C}$ at $x=0.10\ \text{m}$, and $+3.0\ \mu\text{C}$ at $x=0.20\ \text{m}$. Without full numbers, write the vector expression (using Coulomb's law) for the net force on the $+2.0\ \mu\text{C}$ charge and state whether forces add or subtract along x.

Expected: $F_{\text{net}} = F_{\text{from}(-1.0\ \mu\text{C on } +2.0\ \mu\text{C})} + F_{\text{from}(+3.0\ \mu\text{C on } +2.0\ \mu\text{C})}$, where each $F = (1/4\pi\epsilon_0) (q_{\text{source}} q_{\text{test}})/r^2 \hat{r}$ ($\hat{r}$ points from source to test). Along x, forces are collinear so add algebraically with sign: the force from $-1.0\ \mu\text{C}$ is attractive (points toward +x if the source is at $+0.10\ \text{m}$) and the force from $+3.0\ \mu\text{C}$ is repulsive (points toward +x if the source is at $+0.20\ \text{m}$), so include signs accordingly when summing.

Exit Ticket

A tiny test charge $q_{\text{test}} = +2.0 \times 10^{-9}\ \text{C}$ is placed $0.05\ \text{m}$ from a point source charge $q_{\text{source}} = -4.0 \times 10^{-6}\ \text{C}$. (a) Calculate the magnitude and direction of the force on the test charge (show units). (b) In one sentence, state why q_{source} must be an integer multiple of e and whether the total charge of an isolated system can change.

Success indicator: Correct numeric force with units found by $F = (8.99 \times 10^9\ \text{N}\cdot\text{m}^2/\text{C}^2)(|q_1||q_2|)/r^2$ giving the right value and direction (attractive toward the source). One-sentence explanation stating $q_{\text{source}} = n e$ with n an integer (because charge comes from whole electrons/protons) and that total charge in an isolated system is conserved (cannot change).

Homework

- Problem 1 (20 minutes): Calculate the magnitude and direction of the force between $q_1 = -3.0\ \mu\text{C}$ at $x=0$ and $q_2 = +2.0\ \mu\text{C}$ at $x=0.15\ \text{m}$. Show all unit conversions and each algebraic step using Coulomb's law.
- Problem 2 (25 minutes): Three point charges are at the corners of a straight line: $+1.0\ \mu\text{C}$ at $x=0$, $+2.0\ \mu\text{C}$ at $x=0.08\ \text{m}$, and $-1.5\ \mu\text{C}$ at $x=0.18\ \text{m}$. Compute the net force on the middle charge (vector sum along x). Show each pairwise force, signs, and final units in newtons.
- Short answer (10 minutes): A student says rubbing two balloons together 'creates' $+2e$ on one and $-2e$ on the other. Write a one-paragraph (3–5 sentences) correction that uses the word 'transfer' and mentions conservation and integer multiples of e .

Estimated time: 55 min

Mentor Moment — Torsion surprise

In a lab, a student expected rubbing a rod to make some new charge appear; when their tiny torsion balance showed equal and opposite deflections, they realised the rod lost electrons and the pith ball gained them.

Takeaway: Charge isn't created by rubbing — it's moved. Look for equal and opposite changes when you can measure both bodies.

Period 3

Entry Ticket

Write a one-sentence answer on your paper: If a small positive test charge at a point feels a force of 0.020 N to the right, what is the electric field at that point? Give a number, unit, and direction. (No calculator needed.)

Expected response: $E = 0.020 \text{ N/C}$ to the right. Explanation: electric field = force on a +test charge divided by that test charge, here the charge is 1 (so force per unit test charge = 0.020 N/C to the right).

Teacher Script

0-6 min: Quick review and activation

Say: "Class, take out a scrap of paper. Put your name on it. Read the prompt at the top and write one short line with a number, a unit and a direction. Do it now." Pause 30 seconds. Then say: "Who wrote 0.020 N/C to the right? Raise your hand." Point to one student and ask them to read their answer aloud. Then say: "Two quick recall questions: 1) What are the SI units of force and charge?" Wait for responses (N and C). 2) "How does Coulomb force depend on distance between two point charges?" Wait for answer (inverse-square). Conclude: "Good — we will use those facts now to define electric field."

Board: Write: 'Recall: force unit = N ; charge unit = C' and 'Force $1/r^2$ ' under the heading 'Quick review'.

- Anticipate: Why did the entry ticket use N/C instead of just N?
- Anticipate: If the test charge isn't +1 C, how do we know the answer?
- Anticipate: Does the direction 'to the right' mean anything about the source charge?

6-16 min: Derive $E = F/q_{\text{test}}$ from Coulomb's law

Hold up the script and say exactly: "We start with a point source charge Q at the origin and a small positive test charge q_{test} placed a distance r to the right. Using Coulomb's law, the force on q_{test} is $F = (1/4\pi\epsilon_0) Q q_{\text{test}} / r^2$ in the radial direction. Now divide both sides by q_{test} and speak each step as I write it. I will write: $E(r) = F / q_{\text{test}} = (1/4\pi\epsilon_0) Q / r^2 \hat{r}$. Say this with me: 'Electric field is force per unit positive test charge.' Note units: force is N, divide by charge in C, so E has unit N/C. Also show the alternative unit N·m/C when multiplied back by q to get force — but keep N/C as the unit of E . Now read aloud the operational definition from the source: ' $E(r) = \lim_{q \rightarrow 0} F/q$, the force per unit test charge in the limit of negligible test charge.' Emphasize: the test charge must be small so it does not move the source charge."

Board: Write the sourced definitions exactly: 'The vector field produced by a charge that exerts force on other charges placed in the surrounding space.' and ' $E(r) = \lim_{q \rightarrow 0} F/q$, the force per unit test charge in the limit of negligible test charge.' Then write the formula: ' $E = (1/4\pi\epsilon_0) Q / r^2 \hat{r}$ ' and below it 'Units: N/C'.

- Anticipate: Why do we take $q \rightarrow 0$? Will a real test charge ever be zero?
- Anticipate: What is $\hat{r}$ exactly — how do I show its direction?
- Anticipate: Where does $1/4\pi\epsilon_0$ come from? (students may ask about constants)

16-24 min: Demonstration: measure force and compute E

Say: "I will show you a small demo. I have two conducting spheres fixed in place; one is connected to the high-voltage supply and will be our source charge Q . I will bring a tiny probe (a small pith ball on an insulating rod or a calibrated force sensor) to a measured distance r from the source, record the force, and compute $E = F/q_{\text{test}}$. Watch and copy the numbers I call out." Proceed: place probe at $r = 0.20 \text{ m}$ from the center of the charged sphere. Announce: "The probe's charge is

$q_{\text{test}} = +1.0 \mu\text{C} = 1.0 \times 10^{-6} \text{ C}$ (we prepared it). The force sensor reads $F = 0.012 \text{ N}$ away from the source." Write each number as you say it. Then compute aloud step by step: " $E = F / q_{\text{test}} = 0.012 \text{ N} / (1.0 \times 10^{-6} \text{ C}) = 1.2 \times 10^4 \text{ N/C}$ radially away (point right if the probe is on the right)." Ask: "Does this E depend on the fact that the probe's actual q_{test} was $1.0 \mu\text{C}$? If I used $0.5 \mu\text{C}$ and recorded half the force, would E change?" Wait for answers and say: "No — E is independent of q_{test} in the ideal limit."

Board: Write the measurement: 'demo: $r = 0.20 \text{ m}$; $q_{\text{test}} = +1.0 \times 10^{-6} \text{ C}$; measured $F = 0.012 \text{ N}$ ' and the calculation ' $E = 0.012 \text{ N} / 1.0 \times 10^{-6} \text{ C} = 1.2 \times 10^4 \text{ N/C}$ (radially away)'.

- Anticipate: How do we know the probe's charge is exactly $1.0 \mu\text{C}$?
- Anticipate: Won't the probe change the source's charge a bit when we bring it close?
- Anticipate: What if the force sensor reads a negative number — what does that mean?

24-34 min: Guided practice: two worked examples in pairs

Say: "Turn to a partner. We'll do two short calculations together — I'll do the first with you, then you do the second and I'll check. Example A: source $Q = +5.0 \mu\text{C}$, find E at $r = 0.50 \text{ m}$. Steps to say and write aloud with me: convert Q to coulombs $Q = +5.0 \times 10^{-6} \text{ C}$. Compute magnitude using $E = (1/4\pi\epsilon_0) Q / r^2$. Use $1/4\pi\epsilon_0 = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$. So $E = 8.99 \times 10^9 \times 5.0 \times 10^{-6} / (0.50^2) \text{ N/C}$. Calculate: numerator = $8.99 \times 10^9 \times 5.0 \times 10^{-6} = 4.495 \times 10^4 \text{ N}\cdot\text{m}^2/\text{C}$, divide by 0.25 m^2 gives $E = 1.798 \times 10^5 \text{ N/C}$. Direction: because Q is positive, E points radially away from Q (explicitly write $\hat{r}$)." Pause while pairs copy. Then say: "Example B (your turn): $Q = -3.0 \mu\text{C}$ at $r = 0.20 \text{ m}$. Compute magnitude and give direction. I will come around." Circulate and check one pair's work: expected magnitude calculation steps: $Q = -3.0 \times 10^{-6} \text{ C}$, $E = 8.99 \times 10^9 \times 3.0 \times 10^{-6} / (0.20^2) = (8.99 \times 10^9 \times 3.0 \times 10^{-6}) / (0.04) = (2.697 \times 10^4) / 0.04 = 6.7425 \times 10^5 \text{ N/C}$. Direction: towards the charge because Q is negative (write $-\hat{r}$ or 'towards source').

Board: Write Example A full calculation and result: ' $Q = +5.0 \times 10^{-6} \text{ C}$; $r = 0.50 \text{ m}$; $E = 1.80 \times 10^5 \text{ N/C}$, radially away ($\hat{r}$)'. Write Example B set-up: ' $Q = -3.0 \times 10^{-6} \text{ C}$; $r = 0.20 \text{ m}$ ' and leave space for student work. After checking, write the computed result ' $E = 6.74 \times 10^5 \text{ N/C}$, radially toward source ($-\hat{r}$)'.

- Anticipate: Why did we drop the sign when computing magnitude?
- Anticipate: Do we need to include vector arrows on numbers?
- Anticipate: Can we use approximations for 8.99×10^9 like 9.0×10^9 ?

34-40 min: Check, short quiz and summary

Say: "Turn your papers back to me. Two-minute quiz: answer three items on the sheet I'll hand out. Item 1 (concept): If a test charge placed at a point experiences a force opposite to the electric field direction, was the test charge positive or negative? Write one word. Item 2 (numeric): A $+2.0 \times 10^{-6} \text{ C}$ test charge feels a force of 0.040 N toward the left. What is E ? Give number, unit and direction. Item 3 (short): Why must the test charge be 'negligibly small'?" Hand out or display the items and give 2 minutes. Collect answers quickly. Then speak a 30-second summary: read aloud the exact definition from the source once more: 'The vector field produced by a charge that exerts force on other charges placed in the surrounding space.' Remind them of the formula they must remember: $F(r)=q E(r)$ and $E(\text{point}) = (1/4\pi\epsilon_0) q / r^2 \hat{r}$, units N/C . Finish: "Next period we'll add fields from more than one charge using superposition."

Board: Write the quiz items briefly, then write the closing reminders: 'Definition (exact): The vector field produced by a charge that exerts force on other charges placed in the surrounding space.' and ' $F(r)=q E(r)$; $E(\text{point})= (1/4\pi\epsilon_0) q / r^2 \hat{r}$; Units: N/C '.

- Anticipate: On the quiz, for item 2 do we need to divide by a positive number or include sign?
- Anticipate: If the test charge is negative and the force is opposite field, won't that confuse the

direction?

- Anticipate: Will the superposition formula be on the next quiz?

Blackboard Notes

- Quick review
- Definitions
- Point-charge electric field (derived)
- Demo measurement
- Worked examples
- Final reminders
- Recall: force unit = N ; charge unit = C
- Force $1/r^2$ (Coulomb)
- Definition (exact): The vector field produced by a charge that exerts force on other charges placed in the surrounding space.
- Operational definition (exact): $E(r) = \lim_{q \rightarrow 0} F/q$, the force per unit test charge in the limit of negligible test charge.
- Units of E: N/C (force N per charge C)
- Demo: $r = 0.20 \text{ m}$; $q_{\text{test}} = +1.0 \times 10^{-6} \text{ C}$; measured $F = 0.012 \text{ N}$ $E = 1.2 \times 10^4 \text{ N/C}$ (radially away)
- Example A: $Q = +5.0 \text{ } \mu\text{C}$, $r = 0.50 \text{ m}$ $E = 1.80 \times 10^5 \text{ N/C}$, radially away
- Example B: $Q = -3.0 \text{ } \mu\text{C}$, $r = 0.20 \text{ m}$ $E = 6.74 \times 10^5 \text{ N/C}$, radially toward source
- Remember: $F(r) = q E(r)$
- Next: superposition of fields (sum of point-charge fields)
- Diagram: Point source Q at origin and a small test charge q_{test} at distance r on the x-axis; draw $\hat{r}$ arrow pointing away from Q .
- Diagram: Demo setup: large charged sphere, small probe at distance r , arrow showing measured force direction and magnitude label $F = 0.012 \text{ N}$.
- Diagram: For negative Q example: arrow for E pointing toward the source (label 'towards' or $-\hat{r}$).
- Formula: $F(\mathbf{r}) = q \mathbf{E}(\mathbf{r})$
- Formula: $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$
- Formula: $\mathbf{E}(\mathbf{r}) = \sum_i \frac{1}{4\pi\epsilon_0} \frac{q_i}{r_i^2} \hat{r}_i$

Activities

Probe a Charged Rod with a Suspended Test Charge (10 min, demonstration)

Materials: one plastic rod (e.g., PVC or acrylic) or plastic ruler, a piece of wool or cloth for rubbing, one lightweight test probe: small crumpled paper ball (~1 cm) with a tiny piece of aluminum foil taped on, tied to a 20–30 cm length of cotton thread, tape, ruler with millimetre markings, marker, stopwatch or classroom clock (optional for timed steps), whiteboard or chart paper and marker for teacher derivation

- Group students into pairs. Place materials at each pair's table. Say: "We will demonstrate how a source charge produces an electric field, how we define that field operationally, and how to calculate its magnitude and direction for a point charge."
- Demonstration start (0–2 minutes): Rub the plastic rod with the wool to build charge. Hold the rod near (but not touching) the suspended test probe so the probe is free to move. Ask pairs to watch from their seats and note whether the probe is attracted or repelled and in which direction it moves relative to the rod.

- Derive the operational definition (2–5 minutes): On the board, write: electric field E at a point = force on a positive test charge q_0 placed there divided by q_0 , $E = F/q_0$. Explain units: force in newtons (N), charge in coulombs (C), so E has units N/C. While you explain, ask one pair to place their probe 0.10 m from the rod and report the direction of the force on the probe (toward or away from the rod).
- Derive E for a point source charge (5–8 minutes): Using Coulomb's law $F = (1/4\pi\epsilon_0) Q q_0 / r^2$, divide both sides by q_0 to obtain $E(r) = (1/4\pi\epsilon_0) Q / r^2$. Point to each symbol, state its SI unit (Q in C, r in m, E in N/C). On the board compute a worked example with numbers: take $Q = +5.0 \times 10^{-6}$ C, $r = 0.20$ m, and compute magnitude $E = kQ/r^2$ with $k = 8.99 \times 10^9$ N·m²/C²; carry units through every step.
- Class check and wrap-up (8–10 minutes): Ask pairs to compare the observed direction (attraction or repulsion) with the sign of Q in the worked example and state: for positive Q field points away from Q ; for negative Q field points toward Q . While pairs discuss, circulate to confirm that each pair can point to the probe and rod and state the direction and estimate the distance used.

Student instructions:

- Watch the teacher charge the rod and hold it near the suspended test probe. Note whether the probe moves toward or away from the rod and the approximate distance between rod and probe in metres.
- Write the operational definition in this sentence: "The electric field E at a point is the force on a positive test charge q_0 placed there divided by q_0 , so $E = F/q_0$." Include units: force in N, charge in C, so E in N/C.
- Observe the teacher's algebra: start from Coulomb's law $F = (1/4\pi\epsilon_0) Q q_0 / r^2$, divide by q_0 , and copy the result $E(r) = (1/4\pi\epsilon_0) Q / r^2$. Label each symbol with its SI unit on your paper.
- Follow the worked numeric example on the board. Then, using the same numbers, check the direction: if Q is positive, draw an arrow showing the field at the probe location; if Q were negative, draw the arrow pointing the opposite way.
- Answer two quick items on your paper: (a) What is the measured direction of the force on the probe in your observation (toward or away from the rod)? (b) What is the magnitude of E for $Q = +5.0 \times 10^{-6}$ C at $r = 0.20$ m (show units and calculation)?

Success looks like: Each pair can point to the charged rod (source charge) and the suspended probe (test charge) and say whether the probe was attracted or repelled.; Each student copy shows the operational definition $E = F/q_0$ with units N/C written next to E .; At least one member of each pair writes the numeric calculation for E using $Q = 5.0 \times 10^{-6}$ C and $r = 0.20$ m, including units at every step and a final magnitude in N/C.

Support: If you are unsure how to start, copy and complete this sentence frame: "The electric field at a point is $E = F/q_0$. If a test charge $q_0 = 1.0 \times 10^{-6}$ C feels a force $F = 4.5 \times 10^{-3}$ N away from the source, then $E = \underline{\hspace{1cm}}$ N/C." Also, use this worked first item to guide you: $k = 8.99 \times 10^9$ N·m²/C²; for $Q = 5.0 \times 10^{-6}$ C and $r = 0.20$ m, compute $E = kQ/r^2 = (8.99 \times 10^9)(5.0 \times 10^{-6})/(0.20^2) = (8.99 \times 10^9 \times 5.0 \times 10^{-6})/(0.04) = (44.95 \times 10^3)/0.04 = 1.12 \times 10^6$ N/C. Show the units at each multiplication and division step.

Extension: For students who finish early: place two point charges along a line, $Q_1 = +4.0 \times 10^{-6}$ C at $x = 0.10$ m and $Q_2 = -2.0 \times 10^{-6}$ C at $x = 0.30$ m. Calculate the net electric field vector at $x = 0.20$ m along the line. Show vector directions, compute magnitudes from each charge with units, and give the algebraic sum (include sign and final unit N/C).

Checkpoint Questions

- [remember] State the operational definition of the electric field in one sentence and give its SI unit.

Expected: The electric field is the force per unit positive test charge ($E = F/q$) measured in newtons per coulomb (N/C).

- [apply] A point charge $Q = +5.0 \mu\text{C}$ produces a force of 0.045 N on a $+1.0 \mu\text{C}$ test charge at a certain point. Use the operational definition to find the electric field magnitude at that point (show units).

Expected: $E = F/q = 0.045 \text{ N} / 1.0 \times 10^{-6} \text{ C} = 4.5 \times 10^4 \text{ N/C}$. (Units: N/C.)

- [apply] Using $E(r) = (1/4\pi\epsilon_0) Q / r^2 \hat{r}$, indicate the direction of E at distance r from (a) $Q = +3.0 \mu\text{C}$ and (b) $Q = -3.0 \mu\text{C}$. Give a one-line justification for each.

Expected: (a) For $Q = +3.0 \mu\text{C}$, E points radially away from the charge ($\hat{r}$) because a positive test charge is repelled. (b) For $Q = -3.0 \mu\text{C}$, E points radially toward the charge ($-\hat{r}$) because a positive test charge is attracted.

Exit Ticket

Part A (short): Write the operational definition of electric field and its SI unit. Part B (calculation): Compute the magnitude and direction of the electric field at $r = 0.20 \text{ m}$ from a point charge $Q = -2.0 \mu\text{C}$. Show units and state the direction relative to the source charge.

Success indicator: Part A names E as force per unit positive test charge and gives unit N/C. Part B calculates $E = (1/4\pi\epsilon_0) \cdot |Q| / r^2$ numerically (value in N/C with units) and states that the field points toward the negative source charge (radially inward).

Homework

- 1) Compute the electric field magnitude and direction at $r = 0.10 \text{ m}$ and $r = 0.50 \text{ m}$ from a point charge $Q = +4.0 \mu\text{C}$. Show all steps, carry units (N/C) through, and write answers with direction (radially away/toward).
- 2) A small test probe measures a force of 0.012 N when placed 0.15 m from an unknown point charge. If the probe is $+2.0 \mu\text{C}$, find the source charge Q (include sign) using $F = qE$ and E from the point-charge formula. Show units and reasoning.
- 3) Short conceptual: Explain in two sentences why the test charge must be 'negligibly small' when defining the electric field.

Estimated time: 30 min

Mentor Moment — The probe that changed the reading

In a lab, a student used a relatively large test sphere and found different force readings at different probe sizes. Shrinking the probe to a tiny bead removed the discrepancy because the smaller probe disturbed the source charge far less.

Takeaway: Make your test charge small so it measures the field without changing it; always report units (N/C) when you quote E .

Period 4

Entry Ticket

Two point charges $+q$ and $-q$ are fixed $2a$ apart (negative on the left, positive on the right). The pair is placed in a uniform electric field E pointing to the right. In one short sentence, write the net force on the pair and the net torque on the pair (include units).

Expected response: Net force = 0 N (because $+q$ and $-q$ feel equal and opposite forces in a uniform field). Net torque 0 N·m in general: torque tends to rotate the dipole; for example if the line joining charges is at angle θ to E , torque magnitude = $p E \sin\theta$ with units N·m, where p (dipole moment) = $q \cdot 2a$ (units C·m).

Teacher Script

0-5 min: Activate prior knowledge — quick review and worked example

Say: "Quick review: write the electric field from a single point charge q at distance r . Include units."

Pause while one student writes. Then point: "That field has units N/C and depends on $1/r^2$. Now a quick worked example: two equal positive charges $+2.0 \mu\text{C}$ at $x = -0.10 \text{ m}$ and $+2.0 \mu\text{C}$ at $x = +0.10 \text{ m}$. What is the electric field at the origin ($x=0$)?" Ask a volunteer to compute with you on the board. Guide calculation: compute magnitude from each charge using $E = (1/4\pi\epsilon_0) q / r^2$, show numbers: $q = 2.0 \times 10^{-6} \text{ C}$, $r = 0.10 \text{ m}$, $(1/4\pi\epsilon_0) = 9.0 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$. Compute each E in N/C, show directions (both point away from each positive charge) and add vectors by superposition to get net E at origin (they cancel so net $E = 0 \text{ N/C}$). Emphasise units N/C and that we add vectors component-wise."

Board: Write the point-charge field formula and the two-charge superposition sum. Show the numeric substitution and cancellation.

- Anticipate: Why do the fields cancel instead of add? (Because directions are opposite at the origin.)
- Anticipate: Do we need to use components? (Yes when directions differ; here they are opposite on the same line.)
- Anticipate: What value for $1/4\pi\epsilon_0$ should we use? (Use $9.0 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$ for quick estimates.)

5-15 min: Introduce electric dipole and dipole moment; define electric flux

Say: "Now a dipole: take two charges $+q$ and $-q$ separated by distance $2a$. The pair is called an electric dipole. I will write the definition we must use." Read aloud the source definition for dipole moment. Then write $p = q \times 2a$ and say: "This is a vector from the negative charge to the positive charge. Units: C·m." Next: "Write the electric flux definition exactly as given." Write $d\Phi = E \cdot dS = E dS \cos\theta$ and say: "Units for flux are (N/C)·m². dS is a vector whose magnitude is area in m² and whose direction is normal to the surface; for a closed surface dS points outward." Show a quick numeric example: $q = 3.0 \mu\text{C}$, $2a = 0.20 \text{ m}$, compute p magnitude = $q \cdot 2a = (3.0 \times 10^{-6} \text{ C})(0.20 \text{ m}) = 6.0 \times 10^{-7} \text{ C}\cdot\text{m}$ and write the calculation and units on the board."

Board: Write the dipole moment definition, the sample numeric computation for p with units, and the electric flux definition with units.

- Anticipate: Why is p from negative to positive? (This is the chosen convention from the source; it matches the field direction on the axis at far distances.)
- Anticipate: Is p a vector or scalar? (p is a vector; include direction and units C·m.)
- Anticipate: What does flux physically mean? (Proportional to number of field lines crossing;

units (N/C)·m².)

15-23 min: Demonstration: dipole in uniform electric field and computing torque

Set up or draw the apparatus: show two charges +q (right) and -q (left) separated by 2a mounted on a short rigid rod that can rotate, placed in uniform E pointing to the right. Say: "Point out the forces on each charge: $F = qE$, direction depends on sign. Write the two forces as vectors and show they are equal in magnitude and opposite in direction if the dipole axis is perpendicular to E, but generally produce a couple that creates torque." Draw forces on the diagram: on +q force to the right, on -q force to the left. Next compute the torque step-by-step for the general case where the dipole axis makes angle θ with E: write force magnitudes $F = qE$ (units N). Show lever arms: each force acts at distance a from the center, perpendicular component = $a \sin\theta$. Compute torque magnitude from one force as $\tau_{\text{one}} = qE \cdot a \sin\theta$, total torque = $2 \cdot (qE a \sin\theta) = (2 q a) E \sin\theta$. Substitute $p = q \cdot 2a$ and conclude torque magnitude $\tau = p E \sin\theta$ (units N·m). Say: "Vector form: $\tau = p \times E$, direction given by right-hand rule." Walk students through units to show N·m. Emphasise no net force in a uniform field because forces cancel.

Board: Draw dipole in E field with forces labelled ($F = qE$, direction shown), show distances a, label angle θ between p and E, write algebraic steps: $\tau = 2 (qE)(a \sin\theta)$ $\tau = (2 q a) E \sin\theta$ $\tau = p E \sin\theta$, then write vector statement $\tau = p \times E$ and note units N·m.

- Anticipate: Why multiply by 2? (Because both charges produce torques that add.)
- Anticipate: Should we use cross product or scalar magnitude? (We use both: magnitude is $pE \sin\theta$; vector form is $p \times E$.)
- Anticipate: What happens if field is not uniform? (Then forces on the two charges need not cancel — outside today's scope but point out the reason.)

23-33 min: Guided practice in pairs — computations

Give two short problems on the board and ask pairs to work for 6 minutes, then discuss answers: Problem (a): $q = 4.0 \times 10^{-6} \text{ C}$, separation $2a = 0.30 \text{ m}$. Compute p (include units). Problem (b): For the dipole from (a) in a uniform $E = 5.0 \times 10^3 \text{ N/C}$, with the dipole axis at $\theta = 30^\circ$ to E, compute the torque magnitude and state its direction (clockwise or anticlockwise when viewed from above). Circulate, check units and arithmetic. After 6 minutes, select a pair to present their numeric steps: (a) $p = q \cdot 2a = (4.0 \times 10^{-6} \text{ C})(0.30 \text{ m}) = 1.2 \times 10^{-6} \text{ C} \cdot \text{m}$. (b) $\tau = p E \sin\theta = (1.2 \times 10^{-6} \text{ C} \cdot \text{m})(5.0 \times 10^3 \text{ N/C}) \sin 30^\circ = \text{compute: } 1.2 \times 10^{-6} \times 5.0 \times 10^3 = 6.0 \times 10^{-3}$; $\times 0.5 = 3.0 \times 10^{-3} \text{ N} \cdot \text{m}$. Explain direction using right-hand rule: if p points from - to + and angle is 30° above E, torque tries to rotate p toward alignment with E — name clockwise/anticlockwise based on diagram."

Board: Write the two problems, then write the worked numeric solutions with units shown step-by-step for p and τ and label direction of rotation.

- Anticipate: Do we include units in each step? (Yes: show C·m for p and N·m for torque.)
- Anticipate: How to get the sine value? (Use $\sin 30^\circ = 0.5$; calculators OK.)
- Anticipate: How to decide clockwise or anticlockwise? (Use the picture: if p is above E and needs to rotate down, say which direction that is in the diagram.)

33-37 min: Check for understanding — three quick concept checks

Read three short questions aloud and have students write single-line answers on a slip: (1) Does a dipole in a uniform E experience a net force? Answer and units. (2) Why does zero net charge imply no net force in a uniform E? One-sentence explanation. (3) Compute the torque numerically for this quick case: $p = 2.0 \times 10^{-6} \text{ C} \cdot \text{m}$, $E = 1.0 \times 10^4 \text{ N/C}$, $\theta = 90^\circ$ (units). Give 90 seconds for answers, then sample 3 students' answers aloud and correct any errors immediately. Emphasise unit consistency in the numerical answer (N·m)."

Board: Write the three check questions on the board for students to copy, and after sampling, write the correct brief answers and numerical torque with units: $\tau = p E \sin 90^\circ = (2.0 \times 10^{-6})(1.0 \times 10^4)(1) = 2.0 \times 10^{-2} \text{ N}\cdot\text{m}$.

- Anticipate: What if charges are different? (Then not a pure dipole; total charge not zero and a net force can appear.)
- Anticipate: Is the torque calculation different for $\theta = 0$ or 180° ? (Yes: $\sin 0^\circ = 0$ so torque = 0.)
- Anticipate: What if dipole is in a non-uniform field? (We only state that net force can occur then; not computed today.)

37-40 min: Consolidate results and bridge to next period

Summarise aloud while pointing to the board: "We used superposition to combine point-charge fields, defined dipole and dipole moment $p = q \cdot 2a$ (C·m), and defined electric flux $d\Phi = E \cdot dS$ with units (N/C)·m². We derived the torque on a dipole in a uniform E: magnitude $\tau = p E \sin \theta$ and vector form $\tau = p \times E$ with units N·m. Net force in a uniform field is zero for a neutral dipole. Next time we will use p to describe the dipole's far-field behaviour and connect to Gauss's law and Coulomb's law. Make sure your notes include units at every step." End with: "Any final quick questions?" then collect slips or record common errors to address next period.

Board: Underline the final boxed results: $p = q \cdot 2a$ (C·m), $d\Phi = E \cdot dS$ (units), $\tau = p E \sin \theta$ (units N·m) and the note: Net force = 0 in uniform E for neutral dipole.

- Anticipate: Will we need formulas for the dipole field next class? (Yes — we'll derive far-field forms using p.)
- Anticipate: Do we always use $p = q \cdot 2a$ or is there a sign? (p is a vector from negative to positive; magnitude is $q \cdot 2a$.)
- Anticipate: Should we memorise the torque formula? (Understand the derivation and the units; you will use it next time.)

Blackboard Notes

- Point-charge field and superposition
- Electric dipole and dipole moment
- Electric flux
- Torque on dipole in uniform E (derivation)
- Worked examples and key results
- Point-charge field: $E = (1/4\pi\epsilon_0) q / r^2 \cdot \hat{r}$ (units: N/C)
- Superposition: $E(r) = \sum_i (1/4\pi\epsilon_0) q_i / r_i^2 \cdot \hat{r}_i$
- Electric flux (definition): $d\Phi = E \cdot dS = E dS \cos \theta$ (units: (N/C)·m²)
- Electric dipole: two charges +q and -q separated by 2a. Total charge = 0.
- Dipole moment (definition): $p = q \times 2a$ (vector from - to +, units C·m)
- Force on each charge in uniform E: magnitude $F = q E$ (units N)
- Torque derivation (steps):
- torque from one charge = $q E \cdot (a \sin \theta)$ (N·m)
- total $\tau = 2 (q E a \sin \theta) = (2 q a) E \sin \theta$
- substitute $p = 2 q a$ τ (magnitude) = $p E \sin \theta$ (units N·m)
- Vector form (statement): $\tau = p \times E$ (direction by right-hand rule)
- Key point: Net force on neutral dipole in uniform E = 0 N (forces cancel).
- Diagram: Two positive charges example: +2.0 μC at $x = -0.10 \text{ m}$ and +2.0 μC at $x = +0.10 \text{ m}$; arrows showing fields at origin cancel.
- Diagram: Dipole diagram: -q (left) and +q (right) separated by 2a, centre marked, p arrow from - to +, show angle θ between p and uniform E (E arrow to right).

- Diagram: Show forces on each charge: $F_{\text{on}+} = +qE$ (arrow right), $F_{\text{on}-} = -qE$ (arrow left), show lever arms a and perpendicular component $a \sin\theta$, indicate torque sense.
- Formula: $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$
- Formula: $\mathbf{E}(\mathbf{r}) = \sum_i \frac{1}{4\pi\epsilon_0} \frac{q_i}{r_{iP}^2} \hat{r}_{iP}$
- Formula: $d\Phi = \mathbf{E} \cdot d\mathbf{S} = E \, dS \cos\theta$

Activities

Compute torque on a horizontal electric dipole in a uniform field (10 min, simulation)

Materials: One printed 1-page worksheet per pair (teacher can copy in 2 minutes), Pencil and eraser, Calculator, Ruler (cm/mm) and protractor, Whiteboard and marker (teacher use)

1. Arrange students in pairs. Distribute one worksheet and a ruler/protractor to each pair. Tell students: "You have 10 minutes to complete the worksheet. Work together and write numbers with units." Start the timer.
2. Read aloud the physical setup shown on the worksheet: two point charges $+q$ and $-q$ separated by distance $2a$, located on the x -axis, placed in a uniform electric field E pointing at angle θ above $+x$. Say: "We will calculate the dipole moment, the electric field at the midpoint by superposition, the flux through a small square, and the torque $\tau = p \times E$."
3. While pairs work, circulate. For any pair that is stuck, prompt them to (a) identify q in coulombs, $2a$ in metres, and E in newtons per coulomb, (b) draw vectors for p and E on their worksheet, and (c) use the protractor to measure angle between p and E . If all pairs finish early, hand them the extension problem.
4. At 8 minutes, call the class back and ask two pairs to write their numerical answers for p (magnitude with units $C \cdot m$), torque magnitude $|\tau|$ in $N \cdot m$, and statement about net force on the dipole on the whiteboard. Congratulate correct unit-carrying answers and correct qualitative conclusion (net force zero in a uniform E).
5. If time allows (final minute), highlight on the whiteboard the derivation steps: $p = q(2a)$ directed from $-q$ to $+q$ (units $C \cdot m$), field at midpoint $E_{\text{mid}} = E_+ + E_-$ by superposition (show signs), and magnitude $|\tau| = |p| |E| \sin\phi$ where ϕ is angle from p to E (units $N \cdot m$).

Student instructions:

1. Read the setup: two point charges $+q$ and $-q$ are on the x -axis, separated by distance $2a$. The midpoint is at the origin. The uniform external electric field is $E = (E \text{ magnitude}) N \cdot C^{-1}$ at angle θ above $+x$. (All numbers are on your worksheet.)
2. Compute the dipole moment p : multiply q (in coulombs, C) by separation $2a$ (in metres, m). Write p with direction from $-q$ to $+q$ and unit $C \cdot m$. Show p as an arrow on the diagram.
3. Use the superposition principle to find the electric field at the midpoint: compute the field due to $+q$ and $-q$ separately at the midpoint, write each field as a vector with units $N \cdot C^{-1}$, then add them componentwise to get E_{mid} (show arithmetic).
4. Compute electric flux through the small square of side s (given in metres) centered at the midpoint whose normal is $+x$: use $\Phi = E \cdot A = E_x \cdot A$ where $A = s^2$ (m^2). Show units $N \cdot m^2 \cdot C^{-1}$ and state whether flux is positive or negative (based on outward normal).
5. Find the torque on the dipole: draw vectors p and E on the diagram, measure the angle ϕ between them with the protractor, then compute the torque magnitude $|\tau| = |p| |E| \sin\phi$. Write τ as a vector using right-hand rule (direction of $p \times E$) and give units $N \cdot m$. Finally, answer: does a net force act on the dipole in this uniform field? Explain in one sentence.

Success looks like: Each pair shows p calculated with correct units $C \cdot m$ on their worksheet and an arrow drawn from $-q$ to $+q$; Each pair writes component values for the two fields at the midpoint and shows their vector sum

(numbers in $\text{N}\cdot\text{C}^{-1}$) using superposition.; Each pair computes a torque magnitude with correct units $\text{N}\cdot\text{m}$ and states a one-sentence conclusion that net force = 0 in a uniform E .

Support: Sentence frames and worked first item: "Dipole moment $p = q \times 2a = (\text{--- C}) \times (\text{--- m}) = \text{--- C}\cdot\text{m}$ directed from $-q$ to $+q$." Worked example: if $q = 2.0 \times 10^{-6} \text{ C}$ and $2a = 0.10 \text{ m}$, then $p = (2.0 \times 10^{-6} \text{ C})(0.10 \text{ m}) = 2.0 \times 10^{-7} \text{ C}\cdot\text{m}$. For the midpoint field, compute magnitude of each point-charge field using $E = k|q|/r^2$ with $r = a$ (show $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2\cdot\text{C}^{-2}$). Use component addition: $E_{\text{total},x} = E_{+x} + E_{-x}$, $E_{\text{total},y} = E_{+y} + E_{-y}$.

Extension: Replace the uniform field with a non-uniform field that changes with x as $E(x) = E_0 + \alpha x$ (units $\text{N}\cdot\text{C}^{-1}$ where x in m). Ask: for small α (units $\text{N}\cdot\text{C}^{-1}\cdot\text{m}^{-1}$), estimate the net force on the dipole to first order in α , expressing the result in terms of p , α and a . Show units and give a numerical example.

Checkpoint Questions

- [apply] Two point charges $+3.0 \mu\text{C}$ and $-3.0 \mu\text{C}$ are separated by 4.0 cm along the x -axis, with the negative charge at $x = 0$ and the positive at $x = 0.04 \text{ m}$. What is the dipole moment vector p (magnitude, direction, and units)?

Expected: Magnitude $p = q(2a)$ where $q = 3.0 \times 10^{-6} \text{ C}$ and separation $2a = 0.04 \text{ m}$, so $p = 3.0 \times 10^{-6} \text{ C} \times 0.04 \text{ m} = 1.2 \times 10^{-7} \text{ C}\cdot\text{m}$. Direction: from negative to positive charge, i.e. $+x$ direction. So $p = (1.2 \times 10^{-7} \text{ C}\cdot\text{m}) \hat{x}$.

- [apply] A dipole with $p = 2.0 \times 10^{-7} \text{ C}\cdot\text{m}$ lies in a uniform electric field $E = 5.0 \times 10^3 \text{ N/C}$ at angle $\theta = 30^\circ$ between p and E . Calculate the magnitude of the torque τ and give units.

Expected: $\tau = p E \sin\theta$. Compute $\sin 30^\circ = 0.5$. $\tau = (2.0 \times 10^{-7} \text{ C}\cdot\text{m})(5.0 \times 10^3 \text{ N/C})(0.5) = (2.0 \times 10^{-7} \times 5.0 \times 10^3 \times 0.5) \text{ N}\cdot\text{m} = (5.0 \times 10^{-4}) \text{ N}\cdot\text{m}$. So $\tau = 5.0 \times 10^{-4} \text{ N}\cdot\text{m}$.

- [apply] A dipole (total charge zero) is placed in a perfectly uniform electric field. Will the dipole experience a net force? Briefly justify your answer using superposition.

Expected: No net force. Each charge experiences an equal-magnitude force in opposite directions when the field is uniform ($F_+ = qE$, $F_- = -qE$), so vector sum $F_{\text{total}} = qE + (-qE) = 0 \text{ N}$. Superposition of the two point-charge forces gives zero net force.

Exit Ticket

A dipole consists of $+4.0 \mu\text{C}$ and $-4.0 \mu\text{C}$ separated by 6.0 cm , with the negative charge at the origin and the positive at $x = 0.06 \text{ m}$. It sits in a uniform electric field $E = 1.0 \times 10^4 \text{ N/C}$ directed at 60° above the $+x$ axis. Calculate (a) the dipole moment vector p (include units and direction), (b) the torque vector τ (magnitude, direction relative to the page), and (c) whether a net force acts on the dipole (show the reasoning in one sentence).

Success indicator: Student calculates $p = q(2a) = (4.0 \times 10^{-6} \text{ C})(0.06 \text{ m}) = 2.4 \times 10^{-7} \text{ C}\cdot\text{m}$ directed $+x$; computes $\tau = p \times E$ giving magnitude $\tau = pE \sin 60^\circ = (2.4 \times 10^{-7})(1.0 \times 10^4)(3/2) = 2.08 \times 10^{-3} \text{ N}\cdot\text{m}$; states torque direction is out-of-plane (use right-hand rule: into or out of page consistent with $p \times E$) and correctly states net force = 0 N because $+q$ and $-q$ forces cancel in a uniform E .

Homework

- Compute the dipole moment (magnitude and direction) for each: (a) $+2.0 \mu\text{C}$ and $-2.0 \mu\text{C}$ separated by 2.0 cm along y , (b) $+5.0 \mu\text{C}$ and $-5.0 \mu\text{C}$ separated by 10 cm making a 45° line in the xy -plane (show vector components in $\text{C}\cdot\text{m}$).
- For the dipole in (b) above and a uniform field $E = 2.0 \times 10^3 \text{ N/C}$ pointing along $+x$, calculate the torque vector (components and magnitude, include units) and state whether a net force acts. Show the algebraic steps and carry units through every line.
- Short conceptual: Explain in 2–3 sentences why a non-uniform external electric field can

produce a net force on a dipole while a uniform field cannot.

Estimated time: 30 min

Mentor Moment — Lab paddle surprise

In lab, a student placed a charged bead pair in a uniform field expecting it to drift. Instead it only rotated until aligned with the field. They realised equal opposite forces cancel while the off-axis push makes a twist.

Takeaway: When total charge is zero, think 'rotation, not translation' — uniform fields give torque ($\mathbf{p} \times \mathbf{E}$) but no net force.

Period 5

Entry Ticket

Quick pick (2 minutes). For a point charge the electric field magnitude falls as $1/r^2$. For an electric dipole (two equal opposite charges $\pm q$ separated by $2a$) at distances r much larger than a , which of these is the correct dependence of the dipole field magnitude on r ? Circle one: $1/r$, $1/r^2$, $1/r^3$, $1/r^4$. Write the letter and one short reason (one sentence).

Expected response: Circle: $1/r^3$. Short reason: At distances $r \gg a$ the fields from $+q$ and $-q$ mostly cancel their $1/r^2$ pieces and the leading term falls off faster, as $1/r^3$; this is why a dipole's field decreases faster than a single charge's field.

Teacher Script

0-5 min: Warm-up recall and check entry ticket

Collect one-line answers quickly: "Who circled $1/r^3$? Who circled $1/r^2$?" Pause for 3 quick hands. Read one student reason for $1/r^2$ if present. Then say: "We recall three things before we start: (1) Coulomb's law gives force and field units, (2) dipole moment $p = q \cdot (2a/2?)$ — no: we defined p as q times vector from $-$ to $+$ with units coulomb·metre (C·m), and (3) electric flux idea and units. I'll write these on the board now and we will use them with units in every step." If a student says 'I guessed $1/r^2$ ', ask: "What cancellation did you imagine?" and note the answer briefly.

Board: Write short reminders and units.

- Anticipate: Why is dipole moment p defined as q times distance and not q times $2a$? (Answer: p uses the vector from $-$ to $+$; with symmetric charges separated by $2a$, magnitude $p = q \cdot (2a)$.)
- Anticipate: What are the units of flux? (Answer: $\text{N} \cdot \text{m}^2/\text{C}$ or $\text{V} \cdot \text{m}$ depending on route.)
- Anticipate: Is p a vector? (Answer: Yes, direction from negative to positive charge.)

5-17 min: Derive far-field dipole $E \sim 1/r^3$ (axis and equatorial)

Say: "We will do a step-by-step derivation for points on the dipole axis and on the equatorial plane. Keep units at every line. Take charges $+q$ at $x=+a$ and $-q$ at $x=-a$ on x -axis. Point P is at distance r from origin with $r \gg a$. Start with Coulomb's law for a point charge: $E = (1/(4\pi\epsilon_0)) \cdot q/r'^2$ in N/C. First, axis case: P on $+x$ axis at $x=r$. Distances: $r_+ = r-a$ and $r_- = r+a$. Write their magnitudes to first non-zero order using binomial expansion: $1/(r \pm a)^2 = 1/r^2 \cdot (1 \mp 2a/r + 3(a^2/r^2) \dots)$. Show subtraction of contributions from $+q$ and $-q$ and keep leading nonzero term. Explain cancellation of $1/r^2$ terms and that the leading term in E_x is proportional to qa/r^3 . Replace qa by $p/2$? Wait: clarify magnitude: with charges $\pm q$ separated by $2a$, the dipole moment magnitude $p = q \cdot (2a)$. So $qa = p/2$. Substitute to get $E_{\text{axis}} = (1/(4\pi\epsilon_0)) \cdot 2p / r^3$ in direction of p . Do this slowly, writing units: q in C, a in m so p in C·m, $1/(4\pi\epsilon_0)$ in $\text{N} \cdot \text{m}^2/\text{C}^2$, r^3 in m^3 so overall N/C. Repeat for equatorial point at $y=r$ (perp) where contributions add with opposite sign to axis case and show $E_{\text{equatorial}} = -(1/(4\pi\epsilon_0)) \cdot p / r^3$ (pointing opposite to p). Emphasize the signs: axis points along p , equatorial points opposite p . At each algebra step say the units and simplify to N/C."

Board: Write the step-by-step algebra for axis and equatorial expansions, keeping units on key terms.

- Anticipate: Why do $1/r^2$ terms cancel? (Answer: because the charges are equal and opposite; their leading-order fields at large r are equal in magnitude and opposite in sign along that direction.)
- Anticipate: How do you get binomial expansion? (Answer: show $1/(r \pm a)^2 = 1/r^2 \cdot (1 \mp 2a/r + \dots)$, can accept result without full derivation if pressed.)

- Anticipate: Why is q_a replaced by $p/2$? (Answer: because $p = q \cdot (2a)$, so $q_a = p/2$.)
- Anticipate: Are the directions always exactly along or opposite p ? (Answer: on axis yes; equatorial yes for ra approximation.)

17-25 min: Demonstration: torque and fall-off

Set up two small charged spheres on a rotatable jig showing a pair $\pm q$ separated by $2a$. Apply a uniform external E (from two large parallel plates) to show a torque. Say: "Observe rotation: the dipole aligns with E . Measure and call out values: $q = 2.0 \times 10^{-6} \text{ C}$ (if using values), separation $2a = 4.0 \text{ cm}$, so $p = q \cdot (2a) = 2.0 \times 10^{-6} \text{ C} \times 0.04 \text{ m} = 8.0 \times 10^{-8} \text{ C} \cdot \text{m}$. External field E measured as 2000 N/C (or $2.0 \times 10^3 \text{ N/C}$). Compute torque magnitude $\tau = pE \sin\theta$ and units $\text{N} \cdot \text{m}$ right here on the board with numbers. Pick one angle $\theta = 30^\circ$ to calculate a number. Then move the detector probe along the axis away from the dipole and show field decreasing faster than $1/r^2$ (qualitatively) — students note distances (m) and relative readings (N/C). Ask students to write down the measured quantities and units in their notebooks."

Board: Write measured values, compute p numerically with units, compute $\tau = pE \sin\theta$ with numeric result and units.

- Anticipate: Is the external field really uniform? (Answer: approximately uniform between large plates; discuss limits.)
- Anticipate: What if the dipole were free to move? (Answer: it would both experience torque and possibly net force if field non-uniform.)
- Anticipate: How accurate is the fall-off demonstration? (Answer: qualitative; we will rely on algebraic derivation for exact power law.)

25-35 min: Guided practice: compute torque and dipole field numbers

Put students in pairs. Hand out two short problems on the board: (A) Given $p = 6.0 \times 10^{-7} \text{ C} \cdot \text{m}$ and uniform $E = 1.5 \times 10^3 \text{ N/C}$ at angle 45° , compute τ magnitude and units. (B) Given a dipole with $p = 1.0 \times 10^{-6} \text{ C} \cdot \text{m}$, compute the magnitude of E on the axis at $r = 0.50 \text{ m}$ using E_{axis} formula with $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N} \cdot \text{m}^2)$. Work the first problem together on the board step by step, showing units. Then send pairs to work on B; circulate, prompt students to substitute p and r into the formula $E_{\text{axis}} = (1/(4\pi\epsilon_0)) \cdot 2p/r^3$ and check units result N/C . If students get stuck, show algebra of plugging numbers into $1/(4\pi\epsilon_0) \cdot 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$. Encourage unit cancellation: $(\text{N} \cdot \text{m}^2/\text{C}^2) \cdot (\text{C} \cdot \text{m})/\text{m}^3 = \text{N/C}$."

Board: Write both practice problems and then solve (A) with numeric steps and units; keep (B) available for pairs to use and for teacher to complete if asked.

- Anticipate: Can we use $k = 8.99 \times 10^9$ instead of $1/(4\pi\epsilon_0)$? (Answer: yes; it's the same constant with units $\text{N} \cdot \text{m}^2/\text{C}^2$.)
- Anticipate: Why does τ have units $\text{N} \cdot \text{m}$? (Answer: because p ($\text{C} \cdot \text{m}$) times E (N/C) gives $\text{N} \cdot \text{m}$.)
- Anticipate: Why use ra ? (Answer: the far-field formulas assume r much larger than the charge separation so higher-order terms are negligible.)

35-38 min: Formative check (two quick questions)

Hand out or display two short written questions: 1) "State: does an ideal dipole in a uniform external electric field experience a net force? Give a one-sentence explanation and state units of the torque and force." 2) "If the dipole field at r_1 is E_1 , what is the ratio $E(r_1)/E(r_2)$ for $r_2 = 2 \cdot r_1$ using the far-field $1/r^3$ law? Show calculation." Give 2 minutes for answers, collect, mark quickly: for Q1 expect 'no net force; torque may exist; force units N ; torque units $\text{N} \cdot \text{m}$.' For Q2 expect ratio 8. Give immediate verbal feedback and correct common errors.

Board: Write the two questions and the expected short answers after collecting responses.

- Anticipate: Why no net force in uniform E? (Answer: forces on +q and -q are equal and opposite so they sum to zero.)
- Anticipate: What if the field is not uniform? (Answer: then net force can be nonzero; we only say zero for ideal uniform E.)

38-40 min: Consolidation and homework

Read aloud the final boxed formulas on the board and point to units: "These are the far-field dipole formulas and Gauss's law you'll need next. For homework: problems connecting dipole far-field approximations to Gauss-law problems (fields of wires, planes, shells preview)." Remind students to copy formulas and unit-check all answers.

Board: Box the final formulas and write the homework reference (page and problem numbers).

- Anticipate: Do we need to memorize signs for axis vs equatorial? (Answer: yes—axis is $+2p/r^3$ along p; equatorial is $-p/r^3$ opposite p.)
- Anticipate: Will we use Gauss's law next? (Answer: yes, for continuous charge distributions like wires and planes.)

Blackboard Notes

- Quick reminders (units)
- Dipole far-field derivation (axis and equatorial) — keep units
- Demonstration values & torque calculation
- Practice problems
- Key formulas (boxed)
- Gauss's law (reminder)
- Coulomb constant: $k = 1/(4\pi\epsilon_0)$ with units $\text{N}\cdot\text{m}^2/\text{C}^2$
- Dipole moment: $\mathbf{p}$ (vector) = $q \cdot \mathbf{d}$ (C·m). For two charges $\pm q$ separated by $2a$, magnitude $p = q \cdot (2a)$.
- Units to watch: q in C, a and r in m, p in C·m, E in N/C, τ in N·m
- Axis point: P at $x = r$ ($r > a$). Leading nonzero term after cancellation $q a / r^3$ substitute $p = q \cdot (2a)$
- Equatorial point: leading field $-q a / r^3$ substitute p
- Diagram: Sketch: x-axis with $-q$ at $x=-a$, $+q$ at $x=+a$, point P on $+x$ axis at $x=r$. Label distances $r_-=r+a$ and $r_+=r-a$.
- Diagram: Sketch: equatorial point P on y-axis at $y=r$ with same dipole along x. Show field direction opposite to $\mathbf{p}$.
- Diagram: Demonstration set-up: two small spheres separated by $2a$ on a rotatable bar between parallel plates producing E .
- Formula: $\mathbf{E}_{\text{axis}} = \frac{1}{4\pi\epsilon_0} \frac{2\mathbf{p}}{r^3}$
- Formula: $\mathbf{E}_{\text{equatorial}} = -\frac{1}{4\pi\epsilon_0} \frac{\mathbf{p}}{r^3}$
- Formula: $\Phi = \oint_S \mathbf{E} \cdot d\mathbf{S} = \frac{q_{\text{enc}}}{\epsilon_0}$

Activities

Analyze a Small Electric Dipole in a Uniform Field and at Far Distance (12 min, think_pair_share)

Materials: whiteboard or chart paper, marker, stopwatch or visible clock, one printed problem slip per pair (teacher prepares: values for q , a , r , and E)

1. Arrange students in pairs. Hand each pair a printed problem slip that gives a dipole made

of charges $+q$ and $-q$ separated by $2a$, a numeric dipole moment p , a uniform external field E (magnitude and direction), and one far-field distance r where $r \gg a$. Announce: "You will think individually for 2 minutes, discuss with your partner for 4 minutes, then we will share two answers in class for 6 minutes." Start the timer.

- 2. After 2 minutes, signal students to pair up and discuss. While they discuss (next 4 minutes), circulate. For each pair listen for (a) their calculation of torque magnitude and direction using $\tau = p \times E$ with units $\text{N}\cdot\text{m}$, (b) their statement about net force on the dipole in the uniform field with reasoning, and (c) their derivation showing how the dipole field at distance r depends on $1/r^3$ and the sign difference on axis vs equatorial plane using given numeric values to show scale. Prompt pairs who are stuck with: "Show one force pair calculation from Coulomb's law and then combine them to get the dipole far-field scaling."
- 3. After 4 minutes of paired work, call the class together. Ask two different pairs to present: one pair to show their torque calculation step-by-step on the whiteboard (include units), and a second pair to show the short derivation that the far-field dipole potential/field falls as $1/r^3$ and the sign change between on-axis and equatorial plane. Allow 3 minutes for the first presentation and 3 minutes for the second, correcting algebra or unit errors and reminding students to anchor symbols to numbers.
- 4. End by stating aloud the Gauss's law check: ask students to say whether the net flux through a large sphere enclosing the dipole is zero and why, and relate $k = 1/(4\pi\epsilon_0)$ when useful. Collect slips if you will grade them.

Student instructions:

- Think (2 minutes, alone): Read your slip. Identify numeric q (in coulombs), a (in meters), $p = q \cdot (2a) \cdot (\text{direction})$ in $\text{C}\cdot\text{m}$, the uniform field E (in N/C), and the far distance r (m). Calculate: (a) the torque vector $\tau = p \times E$ (show magnitude in $\text{N}\cdot\text{m}$ and direction), (b) state whether a net force acts on the dipole in the uniform field and give one sentence justification using Coulomb's law, and (c) using Coulomb's law for the two charges, expand the field at a point on the dipole axis at distance r (with $r \gg a$) to show how the leading term scales with $1/r^3$; evaluate the numeric size of that field using the given numbers and include units.
- Pair (4 minutes): Compare your calculations with your partner. Together write one neat torque calculation (with units) and one short derivation (3–5 lines) that shows the $1/r^3$ dependence and the sign difference between the on-axis point (along the dipole axis) and an equatorial point (perpendicular to the dipole axis). Use $k = 1/(4\pi\epsilon_0)$ when converting constants to numbers. Be ready to write these on the board.
- Share (when called): One partner will explain the torque calculation step-by-step and the other will explain the far-field derivation and the Gauss's law statement about net flux.

Success looks like: Each pair's written torque calculation shows $\tau = p \times E$ with numeric magnitude in $\text{N}\cdot\text{m}$ and a correct direction (e.g., into or out of the page, given vector directions); Each pair states clearly, in one sentence, that the net force on the dipole in a uniform E -field is zero and cites cancellation of Coulomb forces on the two charges (reference to equal magnitude, opposite direction); At least one pair on the board produces a short algebraic expansion showing the leading dipole field term scales as $1/r^3$ and identifies opposite signs on axis vs equatorial plane, with a numeric evaluation for their given r (units shown).

Support: If you are stuck, use this sentence frame and worked first lines: " $p = q \cdot 2a = (q \text{ value C}) \cdot (2a \text{ value m}) = (\text{compute}) \text{ C}\cdot\text{m}$. $\tau = p \times E$ so $|\tau| = pE \sin\theta = (p \text{ number C}\cdot\text{m}) \cdot (E \text{ number N/C}) \cdot \sin(\theta) = (\text{compute}) \text{ N}\cdot\text{m}$. Direction: use right-hand rule: point fingers along p , curl to E , thumb gives τ direction." For the far-field: start by writing the field from $+q$ at distance $r - a$ $r(1 - a/r)$ and from $-q$ at $r + a$ $r(1 + a/r)$; expand each with first-order binomial approx to get the $1/r^2$ cancellation and the remaining $1/r^3$ term. Pair with a student who completed the example if you

need help.

Extension: Compute the torque and net force if the external field is nonuniform: $E(z) = E_0(1 + \alpha z)$ in the direction of the dipole axis (E_0 in N/C, α in m^{-1}). Derive the net force as a function of α , q , and a , show units N, and evaluate numerically for given α so you can compare its magnitude to the torque found earlier.

Checkpoint Questions

- [apply] A dipole has $p = 4.0 \text{ C}\cdot\text{m}$ pointing in $+x$. It sits in a uniform field $E = 0.50 \text{ N/C}$ in the $+y$ direction. What is the torque vector $\tau = p \times E$ (include units and sign)?

Expected: Compute cross product: $p (4.0 \hat{i}) \times E (0.50 \hat{j}) = (4.0 \times 0.50)(\hat{i} \times \hat{j}) = 2.0 \hat{k}$. So $\tau = 2.0 \text{ N}\cdot\text{m}$ in the $+z$ direction (written $2.0 \hat{k} \text{ N}\cdot\text{m}$).

- [understand] Is there any net force on an ideal electric dipole placed in a strictly uniform electric field? Give one-sentence explanation and state the units you would use for the net force and for torque.

Expected: Net force = 0 because forces on $+q$ and $-q$ are equal in magnitude and opposite, so they cancel. Units: force in newtons (N); torque in newton-meters (N·m).

- [apply] On the axis of a dipole at large distance r the field scales like $1/r^3$. If the field at $r = 1.0 \text{ m}$ is $8.0 \times 10^{-6} \text{ N/C}$, what is the field at $r = 2.0 \text{ m}$ (same axis)? Show the quick calculation and give units.

Expected: Scale by $(1/2)^3 = 1/8$. So $E(2.0 \text{ m}) = 8.0 \times 10^{-6} \text{ N/C} \times 1/8 = 1.0 \times 10^{-6} \text{ N/C}$.

Exit Ticket

A small ideal dipole has dipole moment $p = (2.0 \hat{i} - 3.0 \hat{j}) \text{ C}\cdot\text{m}$. It is placed in a uniform field $E = (0.40 \hat{j} + 0.30 \hat{k}) \text{ N/C}$. (a) Calculate the torque vector $\tau = p \times E$ with units. (b) State whether a net force acts on the dipole and give a one-line justification including units for force.

Success indicator: Correct paper shows (a) the vector cross product $p \times E$ evaluated component-wise to give τ with numerical components and units N·m (for example $\tau = \dots \hat{i} + \dots \hat{j} + \dots \hat{k} \text{ N}\cdot\text{m}$), and (b) a statement 'net force = 0' with a brief reason that forces on the two point charges cancel in a uniform field and the unit of force given as N.

Homework

- Compute the torque (vector and magnitude) for three dipoles: (i) $p = 1.5 \hat{i} \text{ C}\cdot\text{m}$ in $E = 2.0 \hat{j} \text{ N/C}$, (ii) $p = 2.0(\hat{i} + \hat{j}) \text{ C}\cdot\text{m}$ in $E = 1.0(\hat{i} - \hat{j}) \text{ N/C}$, (iii) $p = 0.8 \hat{j} \text{ C}\cdot\text{m}$ in $E = 0.50 \hat{k} \text{ N/C}$. Write units explicitly and show cross-product work. (Estimated time: 15 min)
- Using the far-field axis formula $E_{\text{axis}} = (1/4\pi\epsilon_0)(2p/r^3)$, compute the magnitude of the electric field on-axis for a dipole with $p = 6.0 \times 10^{-6} \text{ C}\cdot\text{m}$ at (a) $r = 0.50 \text{ m}$ and (b) $r = 1.0 \text{ m}$. Use $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$. Carry units through each step and compare the ratio $E(0.5)/E(1.0)$. (Estimated time: 20 min)
- Short explanation (one paragraph): Why does the dipole far-field go like $1/r^3$ instead of $1/r^2$? Anchor your answer to the physical picture of two opposite charges separated by $2a$ and mention what happens as $r \gg a$. (Estimated time: 10 min)

Estimated time: 45 min

Mentor Moment — The lab clamp that taught torque

In my lab a tiny charged bead kept rotating on a pivot until I realised the bead and its opposite partner made a dipole: the pivot didn't move because the net force was zero, but the pair kept turning until p aligned with E .

Takeaway: If a system rotates but its center doesn't move, check torque ($\text{N}\cdot\text{m}$); if the center translates, check net force (N).

Period 6

Entry Ticket

Write a one-line answer (magnitude and direction) to this: A uniformly charged thin spherical shell carries total charge q . What is the electric field at the exact centre of the shell? Give a number or expression and a direction. No explanation needed.

Expected response: $E = 0$ (zero), no direction

Teacher Script

0-5 min: Quick Gauss review & units (activate prior knowledge)

Hold up the sheet titled "Gauss's law" and read it exactly: "Electric flux through a closed surface = $Q_{\text{enclosed}} / \epsilon_0$." Point to ϵ_0 and say: " ϵ_0 is a constant; its units are $\text{C}^2/(\text{N}\cdot\text{m}^2)$ so the right side has units $\text{C}/(\text{C}^2/(\text{N}\cdot\text{m}^2)) = \text{N}\cdot\text{m}^2/\text{C}\cdot\text{m} = \text{N}\cdot\text{m}/\text{C}$. But flux has units $E\cdot\text{area}$, so E must end up in N/C . We will carry units C , m , N/C in every step." Ask: "What is the flux through a closed surface that encloses a single point charge q ?" Wait for reply. Then write the point-charge example quickly: use Coulomb's law for a point charge at center of spherical surface to show how $\text{flux} = E(4\pi r^2) = q/\epsilon_0$ and therefore $E = q/(4\pi\epsilon_0 r^2)$ (we will derive spherical shell formally later).

Board: Write Gauss's law sentence. Underline ϵ_0 and write units. Draw a small point charge at center of a sphere and label q , r . Write the quick derivation: $\text{flux} = E\cdot 4\pi r^2 = q/\epsilon_0$ $E = q/(4\pi\epsilon_0 r^2)$.

- Anticipate: Why do we carry N/C rather than N ? (Because E is force per unit charge.)
- Anticipate: Do we always get E by dividing Q by ϵ_0 ? (Only for suitable symmetry or after computing flux.)
- Anticipate: Why use a sphere here? (Because symmetry makes E constant on the spherical surface.)

5-13 min: Derive field of infinite plane (Gaussian pillbox)

Say: "We now use Gauss's law to find E near an infinite flat sheet with uniform surface charge density σ (σ). Definition: $\sigma = dQ/dS$ with units C/m^2 ." Draw one infinite plane as a horizontal line and place a small pillbox straddling it. Verbally: "Choose a short cylindrical pillbox whose flat faces are equal area A , one face above the plane and one below. The curved side contributes no flux because E is perpendicular to the plane by symmetry." Ask: "What is dQ in terms of σ and A ?" Pause for reply; then state and write: $dQ = \sigma A$ (units C). Now compute flux: $\text{flux} = E\cdot A$ (top) + $E\cdot A$ (bottom) = $2EA$ because E has same magnitude on both sides and points away from the sheet. Apply Gauss's law: $2EA = Q_{\text{enclosed}}/\epsilon_0 = \sigma A/\epsilon_0$. Cancel A and solve for $E = \sigma/(2\epsilon_0)$. Say units explicitly: σ (C/m^2) divided by ϵ_0 ($\text{C}^2/(\text{N}\cdot\text{m}^2)$) gives N/C . Point to a physical example: a charged metal plate; remind that result assumes infinite sheet or near center of a very large sheet."

Board: Draw plane and pillbox. Write definition: $\sigma = dQ/dS$ (units C/m^2). Write $dQ = \sigma A$. Write $\text{flux} = 2EA$. Write Gauss: $2EA = \sigma A / \epsilon_0$ $E = \sigma/(2\epsilon_0)$. Note units N/C beside final line.

- Anticipate: Why is there a 2? (Because flux passes through both faces of the pillbox.)
- Anticipate: What if the plane is a conductor? (Charge sits on surface; outside field is same form but inside conductor $E=0$.)
- Anticipate: How large must a real plate be for 'infinite' to hold? (Enough so edge effects are negligible at the point of interest.)

13-21 min: Derive field of infinite line (cylindrical Gaussian surface)

Say: "Now consider an infinitely long straight wire with linear charge density λ . Definition: $\lambda = dQ/dl$

with units C/m.” Draw a long vertical wire and coaxial cylindrical Gaussian surface of radius r and length L . Ask: “What is the enclosed charge Q_{enclosed} for that cylinder?” Wait briefly, then write: $Q_{\text{enclosed}} = \lambda L$ (units C). Explain symmetry: E must point radially outward and be constant on the curved surface. The flat ends give no flux because E is parallel to them. Compute flux through curved surface: flux = $E (2\pi r L)$. Apply Gauss's law: $E (2\pi r L) = \lambda L / \epsilon_0$. Cancel L and solve: $E = \lambda / (2\pi \epsilon_0 r)$. Explicitly carry units: λ (C/m) divided by ϵ_0 ($\text{C}^2/(\text{N}\cdot\text{m}^2)$) and r (m) gives N/C. Emphasize direction: radially outward (draw arrow). Mention physical example: thin charged wire.

Board: Draw wire and cylindrical surface. Write definition $\lambda = dQ/dl$ (C/m). Write $Q_{\text{enclosed}} = \lambda L$. Write flux = $E (2\pi r L)$. Write Gauss: $E (2\pi r L) = \lambda L / \epsilon_0$ $E = \lambda / (2\pi \epsilon_0 r)$. Add direction: radial ($\hat{r}$). Add units N/C.

- Anticipate: Why ignore the flat ends? (E is parallel to end faces by symmetry, so no flux there.)
- Anticipate: What if wire is not infinite? (Near an end, symmetry breaks and formula no longer exact.)
- Anticipate: Why does E drop as $1/r$ not $1/r^2$? (Because the Gaussian surface area grows linearly with r for a cylinder.)

21-29 min: Derive field of thin spherical shell (inside and outside)

Say: “Finally, a uniformly charged thin spherical shell with total charge q . Use spherical Gaussian surfaces centered on the shell.” Draw sphere and two Gaussian radii: one $r > R$ (outside) and one $r < R$ (inside). State: “The shell’s total charge is q (units C). For $r > R$ by symmetry E is radial and constant on a spherical surface of radius r . Enclosed charge = q . Flux = $E (4\pi r^2)$. Gauss: $E (4\pi r^2) = q/\epsilon_0$ $E = q/(4\pi \epsilon_0 r^2)$, directed radially outward.” Write units N/C. Then $r < R$: “For a Gaussian surface inside the hollow shell the enclosed charge is zero, $Q_{\text{enclosed}} = 0$, so $E = 0$.” Ask students to compare this outside formula to the point-charge result we recalled at the start and confirm they are identical. Emphasize physical meaning: outside the shell it behaves exactly like a point charge at center; inside there is no field from the shell charges.

Board: Draw spherical shell, label radius R and total q (C). Draw two Gaussian radii $r > R$ and $r < R$. Write flux outside: $E (4\pi r^2) = q/\epsilon_0$ $E = q/(4\pi \epsilon_0 r^2)$ (N/C). Write inside: $Q_{\text{enclosed}} = 0$ $E = 0$.

- Anticipate: Why is the inside field zero? (Because no charge is enclosed by a Gaussian surface inside the shell.)
- Anticipate: Is the shell result exact for any shell thickness? (We used a thin shell; a uniformly filled sphere has a different inside E .)
- Anticipate: Does the result depend on how charge is distributed on the shell surface? (If distribution is uniform it holds; asymmetric distribution breaks symmetry.)

29-36 min: Guided practice (teacher-led calculations in pairs)

Hand out a worksheet with three short calculations. Read each aloud and have students work in pairs while you circulate and prompt them to write units at every step.

Problem (a): “An infinite wire has $\lambda = 5 \text{ nC/m}$. Find E at $r = 2.0 \text{ cm}$. Show units.” (Expect use $E = \lambda / (2\pi \epsilon_0 r)$.)

Problem (b): “An infinite plane has $\sigma = 10 \text{ }\mu\text{C/m}^2$. Find E at 1.0 mm from the plane. Show units and direction.” (Expect $E = \sigma / (2 \epsilon_0)$.)

Problem (c): “A thin spherical shell has $q = 2.0 \text{ }\mu\text{C}$. Find E at $r = 0.5 R$ and at $r = 2.0 R$. Show units.” (Expect $E_{\text{inside}} = 0$ and $E_{\text{outside}} = q / (4\pi \epsilon_0 r^2)$.)

Ask pairs to write final answers with numeric substitution and N/C units. Tell them you will collect one answer sheet for quick checking.

Board: Write the three practice problems clearly with given numbers: (a) $\lambda = 5 \text{ nC/m}$, $r = 2.0 \text{ cm}$. (b) $\sigma = 10 \text{ }\mu\text{C/m}^2$, distance 1.0 mm . (c) $q = 2.0 \text{ }\mu\text{C}$, $r = 0.5R$ and $r = 2.0R$. Remind: show units at each step.

- Anticipate: How do I convert nC/m to C/m? (Multiply nC by 10^{-9} ; $5 \text{ nC/m} = 5 \times 10^{-9} \text{ C/m}$.)
- Anticipate: For the plane, why doesn't r appear in the answer? (Because E for infinite plane is independent of distance.)
- Anticipate: When plugging numbers, what value of ϵ_0 do we use? ($\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$; carry units through.)

36-40 min: Check & consolidate (quick quiz and summary)

Project three very short questions and ask students to write answers on a slip: (1) Give σ , λ , ρ definitions with units. (2) True/False: Field outside a thin spherical shell equals field of a point charge at the center. (3) For an infinite line, how does E scale with r ? Collect slips quickly. Then say aloud the common errors: forgetting factors of 2 for the plane, dropping 2π for the wire, forgetting to include units. Finish with the three takeaways: definitions of densities, the three field formulas we derived, and always track units. Dismiss.

Board: Write three consolidation questions for students to copy. Underline common errors: factor of 2, factor of 2π , units. Re-list final takeaway formulas (as written below).

- Anticipate: Will we need these formulas on the test? (Yes, for symmetric charge problems.)
- Anticipate: What if charges are not uniform? (Then Gauss's law still holds but symmetry may not let you solve for E easily.)
- Anticipate: How strict is the infinite assumption? (Infinite is idealized; large compared to observation distance is the practical requirement.)

Blackboard Notes

- Gauss's law (review)
- Surface, linear, volume charge densities
- Infinite plane (pillbox)
- Infinite line (cylindrical surface)
- Thin spherical shell (inside and outside)
- Practice problems
- Gauss's law: flux through closed surface = $Q_{\text{enclosed}} / \epsilon_0$
- ϵ_0 units appear so E has units N/C; carry units through every step
- surface charge density $\sigma = dQ/dS$ (units C/m^2)
- linear charge density $\lambda = dQ/dl$ (units C/m)
- volume charge density $\rho = dQ/dV$ (units C/m^3)
- For an infinite plane: choose pillbox, $dQ = \sigma A$ $2EA = \sigma A / \epsilon_0$ $E = \sigma/(2\epsilon_0)$ (N/C), normal to plane
- For an infinite line: choose coaxial cylinder, $Q_{\text{enclosed}} = \lambda L$ $E(2\pi r L) = \lambda L / \epsilon_0$ $E = \lambda/(2\pi\epsilon_0 r)$ (N/C), radial
- For a thin spherical shell: outside $E(4\pi r^2) = q/\epsilon_0$ $E = q/(4\pi\epsilon_0 r^2)$ (N/C); inside $E = 0$
- Common errors: missing factor 2 for plane, missing 2π for wire, forgetting units
- Diagram: Point charge at center of a sphere (label q and radius r)
- Diagram: Infinite plane with pillbox straddling plane (label area A and arrows for E above and below)
- Diagram: Infinite wire with coaxial cylinder (label λ , radius r , length L)
- Diagram: Thin spherical shell with two Gaussian surfaces: $r > R$ and $r < R$ (label q and R)
- Formula: $\mathbf{E} = \frac{\lambda}{2\pi\epsilon_0 r} \hat{r}$
- Formula: $\mathbf{E} = \frac{\sigma}{\epsilon_0} \hat{n}$
- Formula: $E_{\text{outside}} = \frac{q}{4\pi\epsilon_0 r^2} \hat{r}, \quad E_{\text{inside}} = 0$

Activities

Derive fields from Gauss's Law: plane, wire, shell (8 min, experiment)

Materials: whiteboard or chalkboard, marker or chalk, meter stick or 1 m ruler, calculator (one per pair), printed worksheet with three blank derivation boxes (one per geometry) — 1 per pair, stopwatch or timer

- Group students in pairs. Hand each pair the printed worksheet and a calculator. Say: "In 8 minutes you will use Gauss's law step by step to derive the electric field for three symmetric charge distributions: infinite plane, infinite straight wire, and thin spherical shell. Write each step with units." Start the timer for 8 minutes.
- Direct pairs to work in order: plane first (box A), then wire (box B), then spherical shell (box C). While they work, circulate and watch that each pair labels Gaussian surfaces, writes dQ expressions in terms of the given density (σ , λ , ρ as needed), carries SI units (C/m^2 , C/m , C/m^3) and shows algebraic steps deriving $E(r)$.
- At 3 minutes remaining give an audible prompt: "Finish derivations and be ready to point to the step where you used symmetry and where you computed flux = $E \cdot A$."
- When time is up, ask three pairs (one per geometry) to come to the board and quickly (1 minute each) point to and read aloud: (a) the Gaussian surface used and why, (b) the dQ expression with units, (c) the final formula with units. If any pair did not complete, ask them to underline the missing step on their worksheet.

Student instructions:

- Work with your partner. Use Gauss's law: $E \cdot dA = Q_{\text{enclosed}}/\epsilon_0$ and show every algebra step with units (SI units: ϵ_0 in $C^2 \cdot N^{-1} \cdot m^{-2}$ or equivalently $C^2/(N \cdot m^2)$). Write answers on the worksheet boxes A, B, C.
- Box A — Infinite uniformly charged plane (surface density σ in C/m^2): 1) State the symmetry and choose a Gaussian pillbox. 2) Write $Q_{\text{enclosed}} = \sigma \times \text{area}$ (include units $C/m^2 \times m^2 = C$). 3) Compute flux = $E \times 2A$ (two faces), set equal to $Q_{\text{enclosed}}/\epsilon_0$ and solve for E . Show units for E (N/C or V/m).
- Box B — Infinite straight wire (linear density λ in C/m): 1) State the symmetry and choose a cylindrical Gaussian surface of radius r and length L . 2) Write $Q_{\text{enclosed}} = \lambda \times L$ (units: $C/m \times m = C$). 3) Compute flux through curved surface = $E(2\pi r L)$, set equal to $Q_{\text{enclosed}}/\epsilon_0$ and solve for $E(r)$. Show direction and units for E .
- Box C — Thin uniformly charged spherical shell (total charge Q in C or surface density σ in C/m^2): 1) For r outside the shell: choose a spherical Gaussian surface of radius $r > R$. Write $Q_{\text{enclosed}} = Q$ (units C). Compute flux = $E(4\pi r^2)$ and solve for $E(r)$ showing units. 2) For r inside the shell ($r < R$): explain $Q_{\text{enclosed}} = 0$ and conclude $E=0$ with units.
- At each derivation, explicitly label where superposition or symmetry was used and ensure every dQ expression shows correct SI units.

Success looks like: Each pair has a drawn Gaussian surface labeled (pillbox, cylinder, sphere) for the correct geometry visible on their worksheet.; Pairs have written the Q_{enclosed} expression with correct density symbol and units (e.g., $Q = \sigma \cdot A$ with units $C/m^2 \cdot m^2 = C$; $Q = \lambda \cdot L$ with units C); Each pair reaches and writes the final expression for E with units: $E_{\text{plane}} = \sigma/(2\epsilon_0)$ (N/C), $E_{\text{wire}} = \lambda/(2\pi \epsilon_0 r)$ (N/C) or equivalent, $E_{\text{shell_outside}} = Q/(4\pi \epsilon_0 r^2)$ (N/C), and $E_{\text{shell_inside}} = 0$ (N/C).

Support: If you are stuck, start with this worked example for the plane: Symmetry: field is perpendicular to plane. Gaussian surface: pillbox of face area A (m^2). $Q_{\text{enclosed}} = \sigma \cdot A$ (C). Flux = $E \cdot A$ (top) + $E \cdot A$ (bottom) = $2EA$ (units: $(N/C) \cdot m^2$). Gauss: $2EA = Q_{\text{enclosed}}/\epsilon_0 = \sigma A/\epsilon_0$. Cancel A : $E = \sigma/(2\epsilon_0)$. Units: $[\sigma] = C/m^2$, $[\epsilon_0] = C^2 \cdot N^{-1} \cdot m^{-2}$ so E has units N/C . Pair weaker readers with a stronger reader and ask the stronger reader to read the worksheet steps aloud while the other follows along.

Extension: Derive the electric field on the axis of a finite-length uniformly charged rod of total charge Q and length L at a point distance x from one end. Show all steps beginning from linear charge density $\lambda = Q/L$ (units C/m), set up the integral for dQ , compute the integral, and state the limiting case as $L \rightarrow \infty$ to recover the infinite-wire result. Carry units through the integral and result.

Checkpoint Questions

- [analyze] You place a Gaussian pillbox that straddles an infinite charged plane with surface charge density σ . Write the flux integral and show the algebraic steps that lead to the field magnitude E on one side of the plane. (Answer in symbols, carry units.)

Expected: Flux = $E \cdot dA = E \cdot A$ (top) + $E \cdot A$ (bottom) = $2EA$. Enclosed charge $Q_{\text{enc}} = \sigma \cdot A$ (units C). Gauss's law: $2EA = Q_{\text{enc}}/\epsilon_0 = \sigma A/\epsilon_0$. Cancel A : $E = \sigma/(2\epsilon_0)$. Units: σ is C/m², ϵ_0 is C²/(N·m²) so E in N/C.

- [apply] For an infinitely long straight wire with linear density λ , you choose a cylindrical Gaussian surface of radius r and length L . Write the steps to get $E(r)$. Explain why the curved side contributes and the ends do or do not.

Expected: Symmetry: E is radial and constant on curved surface, and zero on the flat end faces because E is perpendicular to their normals. Flux = $E \cdot (\text{curved area}) = E(2\pi rL)$. $Q_{\text{enc}} = \lambda L$. Gauss's law: $E(2\pi rL) = \lambda L/\epsilon_0$ $E = \lambda/(2\pi \epsilon_0 r)$. Units: λ is C/m so E in N/C.

- [analyze] State the electric field produced by a uniformly charged thin spherical shell at a point r inside the shell ($r < R$) and at a point outside ($r > R$). Give a one-sentence reason, using Gauss's law, for the inside result.

Expected: Outside ($r > R$): $E = q/(4\pi \epsilon_0 r^2)$ radial (N/C). Inside ($r < R$): $E = 0$ (N/C). Reason: For a Gaussian sphere of radius $r < R$ no charge is enclosed ($Q_{\text{enc}} = 0$) so $E \cdot dA = 0$ $E = 0$ by symmetry.

Exit Ticket

A large metal plate has $\sigma = 10.0 \mu\text{C}/\text{m}^2$. A long thin wire parallel to the plate carries $\lambda = 5.0 \text{ nC}/\text{m}$ and is located 2.0 cm from the plate. Treat the plate as an infinite plane and the wire as an infinite line. Using superposition, write expressions (with units) for the magnitudes of the electric fields from the plate and from the wire at the point halfway between them (1.0 cm from the plate). Then state the net field direction assuming the plate is positively charged and the wire is positively charged.

Success indicator: Student writes $E_{\text{plate}} = \sigma/(2\epsilon_0)$ with $\sigma = 10.0 \times 10^{-6} \text{ C}/\text{m}^2$ giving numerical N/C; $E_{\text{wire}} = \lambda/(2\pi \epsilon_0 r)$ with $\lambda = 5.0 \times 10^{-9} \text{ C}/\text{m}$ and $r = 0.01 \text{ m}$ giving numerical N/C; then states net field magnitude as vector sum of magnitudes with directions: plate field points away from plate, wire field points radially away from wire, and specifies the resulting direction at the midway point.

Homework

- Calculate the electric field magnitude and direction 2.0 cm from an infinite wire with $\lambda = 5.0 \text{ nC}/\text{m}$. Show all steps, include units at each line, and give a numeric answer in N/C.
- A uniformly charged infinite plane has $\sigma = 10 \mu\text{C}/\text{m}^2$. Find the electric field 1.0 mm from the plane. Show units through the algebra and give the numeric N/C value.
- A thin spherical shell has total charge $q = 2.0 \mu\text{C}$ and radius $R = 5.0 \text{ cm}$. (a) Compute E at $r = 3.0 \text{ cm}$ (inside) and $r = 10.0 \text{ cm}$ (outside). Show Gauss's law steps and units, and give numeric answers in N/C.

Estimated time: 30 min

Mentor Moment — The Van de Graaff Plate Surprise

In a demo, a charged metal plate made hair stand on end for students nearby. They expected the field to drop with distance but hair reacted even a few centimetres away because a charged plate's field stays nearly constant close to it.

Takeaway: For a large charged plate the field near it is roughly constant and equals $\sigma/(2\epsilon_0)$, so moving a short distance doesn't change the field much.

Learning Gaps & Misconceptions

[high] If I double the distance between two charges, the electrostatic force becomes half of what it was.

- Diagnostic: Two identical point charges separated by 1.0 m repel with force F . If the separation is increased to 2.0 m, what is the new magnitude of the force?

Reveals: Whether the student recognises the inverse-square dependence (correct: $F/4$) or instead applies an inverse-linear rule (wrong: $F/2$).

- Diagnostic: Two point charges separated by 3.0 m have force F between them. If the separation is reduced to 1.0 m, what is the new force?

Reveals: Whether the student computes the factor ($3^2 = 9$ force increases by 9) or uses linear scaling (factor of 3). A linear answer shows they are treating dependence as inverse-linear, not inverse-square.

- Remediation: Begin with a hands-on lab: measure electrostatic force (or use a Coulomb-force simulation) for one fixed pair of point-like charges at separations 1.0 m, 2.0 m and 3.0 m. Have students record the numerical forces and compute the ratios $F(1\text{m})/F(2\text{m})$ and $F(1\text{m})/F(3\text{m})$.

Why: Concrete measurements force students to confront the actual numeric change. Seeing ratios near 4 and 9 (not 2 and 3) ties the exponent to measurable outcomes and breaks the habit of linear scaling.

- Remediation: Guide a derivation on the board: start from the statement 'force $\propto 1/r^2$ ', substitute $r/2$ and $r/3$, and carry units and algebra through to show F becomes $F/4$ and $F/9$. Ask students to explain each algebraic step in plain language (e.g. 'squaring the distance doubles squaring effect').

Why: Deriving the numerical factor step-by-step links the algebraic form to the numeric change; requiring them to paraphrase prevents memorised symbol manipulation and shows why the exponent matters, not just the word 'inverse'.

- Remediation: Use contrasting examples: compare three laws (inverse, inverse-square, inverse-cube) on a table with sample r values (1,2,3) and computed factors, and ask students to predict which physical situations match each law (e.g. springs $\sim$ linear, gravity/electrostatics $\sim$ inverse-square). Follow with a short quiz question that asks for the factor when r is scaled by arbitrary k .

Why: Placing inverse-square alongside other dependences highlights the qualitative difference produced by changing the exponent. Connecting to other physical systems provides a source of correct generalisation and helps students replace the incorrect rule-of-thumb with a discriminating strategy for interpreting 'varies as'.

[high] If the total charge of an object or system is zero, it produces no electric field anywhere (so neutral means no field).

- Diagnostic: Two point charges, $+Q$ and $-Q$, are fixed a short distance d apart (a dipole). At a point along the line perpendicular to the midpoint and at distance R (with $R \gg d$), is there an electric field? If so, is it zero, and what direction does it point?

Reveals: If a student answers 'zero' this reveals they believe net charge alone determines whether a field exists and are not applying vector superposition to spatially separated charges.

- Diagnostic: A neutral insulating rod has $+Q$ concentrated near one end and $-Q$ concentrated near the other. A small positive test charge is brought near the $+Q$ end. Will the test charge feel a force? If yes, in which direction?

Reveals: If a student says 'no force', it shows they think neutrality eliminates local fields and forces, ignoring local

nonzero fields from charge separation.

- Remediation: Start with a live or simulated demonstration: place two equal and opposite point charges separated by a small distance. Use an electric field sensor, a compass-like field indicator, or a field-line simulation to map the field at several nearby points and at a point far away. Record vector directions and magnitudes and display numeric sensor readings with units (N/C).

Why: Seeing nonzero, directional field readings despite net zero charge confronts the overgeneralisation from neutrality. Explicit numeric vector measurements force students to treat field as a vector sum of contributions from each charge (not a single-scalar property of 'net charge'). Carrying units (N/C) ties the abstract to measurable force per unit test charge.

- Remediation: Derive the field of the dipole on the board step-by-step: write the expression for the field from each point charge (showing magnitudes and directions with units), expand for $R \gg d$ to show the leading-order nonzero term, and compare the $1/r$ dependence of a single charge vs. the faster fall-off for the dipole. Keep units through every line and compute a numerical example (choose Q , d , R) so students calculate concrete N/C values.

Why: A careful algebraic superposition shows cancellation is partial, not complete, and the derivation anchors the result in the same arithmetic students use in other problems. The numerical example connects the symbolic expansion to measurable magnitudes so students stop treating 'zero net charge' as a rule and instead see how geometry controls the field.

- Remediation: Give a short guided worksheet: several quick tasks where students must predict field direction and relative magnitude (then check with simulation/sensor): (a) two like charges, (b) two opposite charges separated, (c) neutral distribution with separated clusters. Each task asks them to compute vector sums for at least two points and to explain in one sentence why 'net charge = 0' does or does not imply zero field at that point.

Why: Active calculation and explanation forces students to practise vector addition and to articulate the specific failure of the misconception (distribution matters). Having them compare prediction with measurement/simulation gives immediate feedback that contradicts the false generalisation, and the explanation step promotes conceptual change rather than memorising a correction.

[high] If there are several charges near a test charge, each charge's force on the test charge is different from what it would be if the other charges weren't there — so you have to 'adjust' each pairwise force to account for the others.

- Diagnostic: Three point charges lie on a line: a $+2 \mu\text{C}$ at $x = 0 \text{ m}$, a $+1 \mu\text{C}$ at $x = 0.10 \text{ m}$, and a $+3 \mu\text{C}$ at $x = 0.30 \text{ m}$. What is the force on the $+1 \mu\text{C}$ charge? (A student who holds the misconception will answer incorrectly.)

Reveals: If the student changes the computed force between $+1 \mu\text{C}$ and one neighbour because of the presence of the other, they do not apply superposition correctly.

- Diagnostic: You measure the force on a small test charge q_0 when only charge Q_1 is present and get F_1 . Then you remove Q_1 and instead put only Q_2 in the same place where Q_1 had been and measure force F_2 . Now you put both Q_1 and Q_2 together; what do you expect the force on q_0 to be?

Reveals: A wrong answer shows the student thinks the presence of the second charge changes the pairwise interaction between the first and the test charge (not simply that two forces exist and sum).

- Remediation: Lab or live demo: use a Coulomb-force sensor or high-quality simulation. Place a test charge q_0 on a force sensor and measure the force when only Q_1 is present; record

magnitude and direction with units (N). Remove Q1 and measure force with only Q2 present. Finally place Q1 and Q2 together and measure the force on q0. Have students compute the vector sum $F_1 + F_2$ (show units N) before revealing the measured combined force.

Why: Concrete measurement ties the abstract statement to physical evidence and units. Seeing that the measured combined force equals the arithmetic vector sum of the two previously measured forces shows that the presence of another charge does not alter the individual pairwise forces on the test charge.

- Remediation: Derivation and guided worksheet: derive the force on a test charge from Coulomb's law for two source charges by writing the forces from each source explicitly as vectors (showing magnitudes in N and unit vectors), then add them algebraically to get the total. Emphasise step-by-step where linearity enters (no cross-terms). Then repeat the algebra including an extra charge to show the pattern generalises.

Why: A stepwise derivation makes it clear that Coulomb's law gives independent pairwise forces that sum; showing units (N) and unit vectors prevents hiding the vector nature. Demonstrating there are no terms that multiply charges together beyond pairwise products addresses the mistaken idea that charges 'modify' other pairwise forces.

- Remediation: Conceptual question sequence and analogy: ask students to consider gravitational forces between small masses as a close analogy — have them calculate the gravitational force on a small mass due to two separated masses by summing vectors, then discuss why adding a third object doesn't change the force between any specific pair. Follow with targeted questions: 'If Q2 is far away, does it change the Coulomb force between Q1 and q0?' and ask for justification referring to the measurements and algebra.

Why: Analogies to familiar inverse-square forces where superposition holds help students transfer correct intuition. The guided questions force explicit reasoning about independence of pairwise interactions rather than vague appeals to 'influence' or 'shielding.'

[high] There is no electric field at a point until I put a test charge there, and the field depends on which test charge I use (its size or sign).

- Diagnostic: You remove every charged object except one point charge +Q fixed in space. At a location 10 cm from +Q there is no test charge present. Is there an electric field at that location? Explain and, if yes, state how its magnitude depends on anything you place there later.

Reveals: Whether the student thinks the field is only present when a test charge is placed there, or whether they understand the field exists independently. A wrong answer shows they conflate force measurement with existence of the field.

- Diagnostic: A point charge +Q is fixed. You first place a small test charge +q1 at a point and measure a force F1; later you replace it with a different test charge +q2 and measure force F2. If $q_2 = 3 q_1$, what should you expect for F2 and for the electric field magnitude E at that point in the two trials?

Reveals: Whether the student recognises that force changes proportionally with test charge while the field (F/q) stays the same. A wrong answer shows they believe the field itself changes with the test charge.

- Remediation: Derive $E = F/q$ at the board starting from Coulomb's law: write $F = k Q q / r^2$, then divide both sides by q to get $E = k Q / r^2$. Work the algebra symbolically and then plug in two different numerical q values to show E is unchanged.

Why: Algebraically cancelling q directly addresses the source of the mistake (confusion from the product form of Coulomb's law). Showing the same E with different numerical q makes explicit that the probe's charge drops out and that E depends only on source charges and geometry.

- Remediation: Class experiment or computer simulation: fix a source charge Q and measure the force on two different test charges (one positive small q_1 , one larger q_2 , and one negative $-q_3$). Students compute $E = F/q$ for each trial and record the values.

Why: Hands-on measurement ties the abstract quantity E to measurable numbers while revealing the pattern: forces change with q but the quotient F/q stays the same (and flips sign appropriately for negative q). This confronts the 'nothing exists until probed' intuition by showing consistent E values even when different probes are used.

- Remediation: Show a field-line diagram or a vector-field plot for the same source charge with no test charge drawn. Ask students to sketch or describe what a small positive and a small negative test charge placed at a point would do, predicting directions and relative magnitudes, then compare to $E = kQ/r^2$.

Why: Separates the visual model of a field (arrows in space) from specific probes and reinforces that the field is a property of space around sources. Asking students to predict probe behaviour from that field challenges the reverse inference (that probes create the field).

[medium] Rubbing two objects creates new electric charge — when I rub a rod on fur (or two materials together) I make charge out of nothing so the total amount of charge in the system increases.

- Diagnostic: You rub a plastic rod on your wool sweater. Before rubbing, the rod and sweater together had zero net charge. After rubbing, the rod is negatively charged and the sweater is positively charged. Has the total electric charge of the rod+sweater system changed? Explain.

Reveals: Whether the student believes rubbing created net charge rather than transferring it between bodies.

- Diagnostic: You take an isolated box containing two neutral objects. You rub them together inside the closed box. After rubbing, you measure each object and find one is negative and the other positive. What is the net charge inside the box now, and why?

Reveals: If the student insists that rubbing makes new charge appear inside an isolated system (violating conservation) instead of recognising equal and opposite transfer.

- Remediation: Token-transfer activity (10 minutes): Give each pair of students a set of small identical tokens labelled 'electron' and two cardboard cut-outs representing Material A and Material B. Start both materials neutral (same number of electron tokens). Have students physically move a small, fixed number of electron tokens from A to B to model rubbing, then count tokens on each.

Why: A concrete particle model makes 'transfer' visible: students see no tokens are created or destroyed, only moved. This addresses the root overgeneralisation by showing how local charging arises from redistribution of discrete charges.

- Remediation: Quantitative classroom demo with an electroscope or simple Faraday-cup setup (15 minutes): Show the electroscope reading when two neutral metal cans are connected (zero). Rub a plastic rod on fur to charge it, touch the rod to one can (observe electroscope change), then touch the two cans together and show the combined reading equals the algebraic sum of earlier readings (or that the net charge of the closed system before and after remains the same). If available, use an electrometer to measure total charge before and after rubbing two objects inside an isolated container.

Why: A measured, repeatable experiment ties the abstract conservation law to numbers and instrument readings. Seeing that instrument readings add algebraically (and that the total measured charge of an isolated system stays constant) corrects the intuition that rubbing 'creates' charge.

- Remediation: Guided questioning and atomic picture (8 minutes): Lead students through a short sequence: (a) Ask what a material's neutral state means in terms of electrons/protons; (b) Ask where extra negative charge on a rubbed object comes from (answer: electrons moved from the other object); (c) Ask what must happen to the other object's charge when electrons leave. Use unit notation (charge in coulombs) for an example transfer (e.g. -3.2×10^{-19} C corresponds to one electron) to show discrete transfers and conservation.

Why: Explicitly connecting macroscopic observations to the microscopic electron-transfer model and carrying units/quantities enforces that charge is quantised and conserved. Using a specific numerical example prevents hand-waving and helps students replace 'creation' language with 'transfer of discrete charges.'

[medium] Electric charge placed on any material always stays where it is put and does not move.

- Diagnostic: You touch a charged rod to one point on a metal (conducting) sphere and then remove the rod. Where is the net charge located on the sphere after you remove the rod?

Reveals: If the student says the charge remains only at the point where the rod touched, they still believe charge is immobile on conductors and have not grasped free electron redistribution.

- Diagnostic: You charge a small patch on a plastic ruler by rubbing it with a cloth. Then you do the same rubbing to one patch on a metal spoon (both isolated on insulating stands). After removing the cloth, in which case does the charge stay only at the rubbed patch and in which case does it spread out?

Reveals: If the student says the charge stays localized on both the ruler and the spoon, they are overgeneralising the insulator behaviour to conductors and haven't learned that conductors let charge move to the surface.

- Remediation: Prediction prompt: Ask students to predict what will happen when a charged metal sphere A is touched to an identical neutral metal sphere B (both on insulating stands). Record predictions. Then perform the classic experiment: charge sphere A (by rubbing or using an electroscope), touch A to B briefly, separate them, and measure the charge on each with an electroscope or by observing attraction to small bits of paper.

Why: Making predictions forces commitment to the erroneous rule and makes the subsequent observable contradiction meaningful. Demonstrating that both spheres share/redistribute the charge provides direct, measurable evidence that charge moves across conductors rather than staying where placed.

- Remediation: Faraday pail or metal-can demo: Place a charged conductor inside a metal pail connected to an electroscope (or use a metal can as a conductor). Show that the electroscope registers the same net charge regardless of where the charge is placed on the inner object; then touch the inside and outside to show charge resides on the outer surface of the conductor.

Why: This ties the abstract rule (free charges move to the surface of a conductor) to a macroscopic measurement. Students observe that charge redistributes to the conductor's surface and that internal points do not retain isolated charge, countering the 'charge stays where put' generalisation.

- Remediation: Direct contrast activity: Have pairs of students charge a small plastic strip and a small metal strip in identical ways. Let them mark the rubbed spot and then probe different locations (with electroscope, tiny pith balls, or by measuring attraction of small paper) to map where the charge effects occur on each material. Students then compare results and explain differences.

Why: Confronting the two materials side-by-side links the observed difference to the cause named in the misconception: material-dependent electron mobility. Students see the localized effect on the insulator and the redistributed effect on the conductor and must reconcile their prior overgeneralisation with this evidence.